Continious - Time : CT Sürekli sinyal Analog

Discrete - Time : DT Ayrık Sinyal Dijital

$x(t) \rightarrow$ [H] $\rightarrow y(t)$

↓ karakteristik

$y(t) = H_x(t) = 2 \cdot x(t)$

↳ çıkışı oluşturmak için girişi 2 ile çarp

↳ x bir fonksiyondur H de bir fonksiyondur

$x \rightarrow [H_1] \xrightarrow{H_1 x} [H_2] \rightarrow y \qquad y = H_2[H_1(x)]$

Çift Sinyal

$x(t) = x(-t)$

$x[n] = x[-n]$

$x_e(t)$

Tek Sinyal

$x(t) = -x(-t)$

$x[n] = -x[-n]$

$x_o(t)$

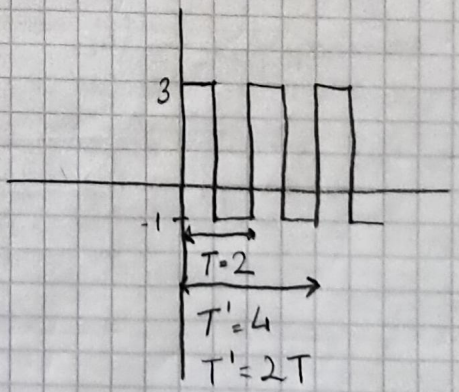
$x(t) = x_e(t) + x_o(t)$

$f : H_z = \frac{1}{s}$

$T = \frac{1}{s}$: periyot

$f = \frac{1}{T}$ $\omega = 2\pi \cdot f$

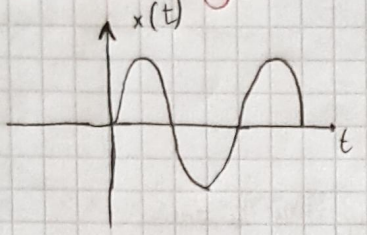
$\omega = \frac{2\pi}{T}$



Sinyal nedir?

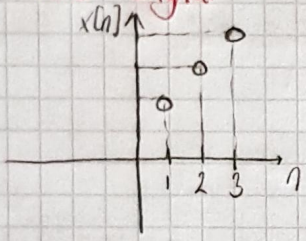
Fiziksel değişkenlerin durumu hakkında bilgi taşıyan ve matematiksel olarak fonksiyon biçiminde ifade edilen kavramdır.

Continious Time Signal



$t \in R$

Discrete Time Signal

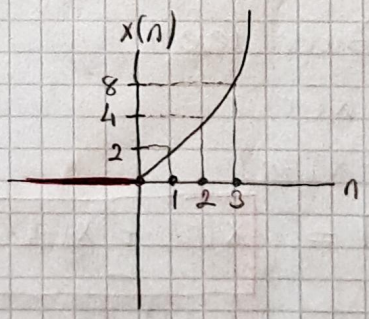


$x[n] \rightarrow n \in Z$

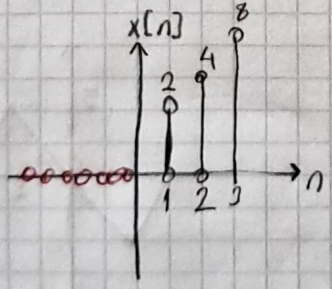
Örnek: 2^n fonksiyonunu $n>0$ olmak üzere D.T ve C.T olarak çiziniz

C.T

$n=0$	$x(n)=0$
$n=1$	$x(n)=2$
$n=2$	$x(n)=4$
$n=3$	$x(n)=8$



D.T



x(n) → açık parantez var ise continious signal

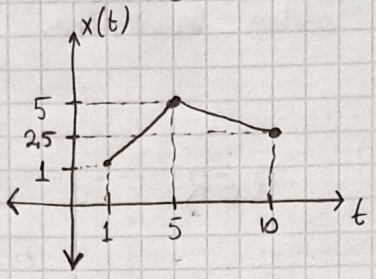
kapalı parantez var ise discrete time signal

Örnek:

$$x(t) = \begin{cases} t & 0 \le t \le 5 \\ \frac{25}{t} & 5 \le t \le 10 \\ 0 & \text{otherwise} \end{cases}$$

$x(t)$ ✓ continious time signal

sinyalini çiziniz

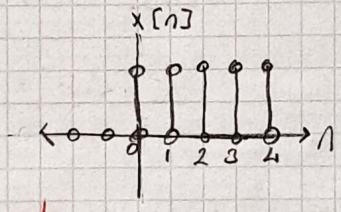


Örnek:

$$x[n] = \begin{cases} 1 & n \ge 0 \\ 0 & n < 0 \end{cases}$$

$x[n]$ ✓ discrete time signal

Sinyalini çiziniz.



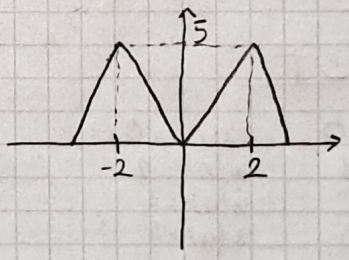
Tek ve Çift Fonksiyonlar

Çift fonksiyonda (even)

$$x(t) = x(-t)$$

$$x[n] = x[-n]$$

$$\text{Even } |x(t)| = \frac{x(t) + x(-t)}{2}$$

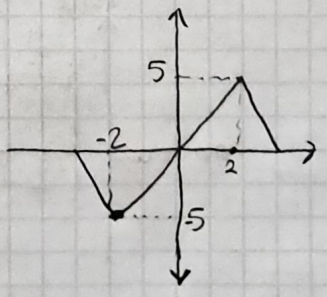


tek fonksiyonda (odd)

$$x(t) = -x(-t)$$

$$x[n] = -x[-n]$$

$$\text{Odd } |x(t)| = \frac{x(t) - x(-t)}{2}$$



Örnek: $x(t) = \cos t + \sin t + \cos t \sin t$ fonksiyonunun çift ve tek kısımlarını bulunuz.

$x(t) = -x(-t)$ → sin(t)

$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$\Rightarrow \frac{\cos t + \sin t + \cos t \sin t + (\cos t - \sin t - \cos t \sin t)}{2}$$

$$x_e(t) = \frac{2\cos t}{2} \Rightarrow x_e(t) = \cos t$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

$$\Rightarrow \frac{\cos t + \sin t + \cos t \sin t - (\cos t - \sin t - \cos t \sin t)}{2}$$

$$\Rightarrow \frac{2\sin t + 2\cos t \sin t}{2} = \sin t + \cos t \sin t$$

Örnek: $x(t) = u(t)$

1, $t \geq 0$

0, $t < 0$

$$x_e(t) = \frac{u(t) + u(-t)}{2}$$

$t > 0 \rightarrow \frac{1}{2}$

$t < 0 \rightarrow \frac{1}{2}$

$$\Rightarrow \frac{1 + 0}{2} = \frac{1}{2}$$

$$x_o(t) = \frac{u(t) - u(-t)}{2}$$

$t > 0 \rightarrow \frac{1}{2}$

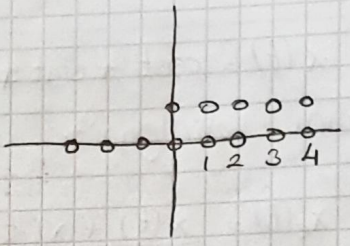
$t < 0 \rightarrow -\frac{1}{2}$

$$\Rightarrow \frac{1 - 0}{2} = \frac{1}{2}$$

$$\frac{0 - 1}{2} = -\frac{1}{2}$$

Örnek:

$$x[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \Rightarrow$$



$$x_e[n] = \frac{x[n] + x[-n]}{2}$$

$n>0$ için $x_e[n] = \frac{x[1] + x[-1]}{2} = \frac{1}{2}$

$n=0$ için $x_e[n] = \frac{x[0] + x[-0]}{2} = 1$

$n<0$ için $x_e[n] \frac{x[-1] + x[1]}{2} = \frac{1}{2}$

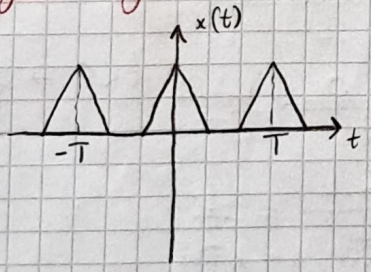
$$x_o[n] = \frac{x[n] - x[-n]}{2}$$

$n>0$ için $x_o[n] = \frac{x[1] - x[-1]}{2} = \frac{1}{2}$

$n=0$ için $x_o[n] = \frac{x[0] - x[-0]}{2} = \frac{0}{2} = 0$

$n<0$ için $x_o[n] = \frac{x[-1] - x[1]}{2} = \frac{-1}{2}$

Periyodik Sinyaller



$x(t) = x(t+T)$

$x[N] = x[n+N]$

Özel durum

$e^{jxt} \Rightarrow x = \omega_0$

$\omega_0 = \frac{2\pi}{T}$ veya $2\pi \cdot f$

$\sin(\alpha t) \Rightarrow \alpha = \omega_0$

Örnek: $x(t) = j \cdot e^{j5t}$ fonksiyonunun temel periyodu nedir?

$\omega_0 = 5 \quad \omega_0 = \frac{2\pi}{T} \quad \frac{2\pi}{T} = 5 \quad 5T = 2\pi$

$T = \frac{2\pi}{5}$

Örnek: $x(t) = \sin(50\pi t)$ fonksiyonunun temel periyodu nedir?

$\omega_0 = 50\pi$

$\omega_0 = \frac{2\pi}{T}$

$\frac{2\pi}{T} = 50\pi$ $\quad T = \frac{1}{25}$

Örnek: $x(t) = 2.\cos(10\pi t + \frac{\pi}{6})$ fonksiyonunun temel periyodu nedir? ($\frac{\pi}{6}$ → ilgilendirmez)

$\omega_0 = \frac{2\pi}{T}$

$\omega_0 = 10\pi$

$\frac{2\pi}{T} = 10\pi$

$T = \frac{1}{5}$

Örnek: $x(t) = \underbrace{2.\cos(10t+1)}_{T_1} - \underbrace{2.\sin(4t-1)}_{T_2}$ fonksiyonunun temel periyodu nedir?

$\frac{T_1}{T_2}$ sonucu rasyonel bir sayı ise fonksiyon periyodiktir.

$T_1 = 2.\cos(10t+1)$

$\omega_0 = 10$

$\frac{2\pi}{T} = 10$

$10T = 2\pi \Rightarrow T = \frac{\pi}{5}$

$T_2 = 2.\sin(4t+1)$

$\omega_0 = 4$

$\frac{2\pi}{T} = 4$

$4T = 2\pi \Rightarrow T = \frac{\pi}{2}$

$\frac{T_1}{T_2} = \frac{\frac{\pi}{5}}{\frac{\pi}{2}} = \frac{2}{5}$ ✓ fonksiyon periyodiktir.

$5T_1 = 2T_2$ $\quad 5.\frac{\pi}{5} = 2.\frac{\pi}{2}$ $\quad T = \pi$

Örnek: $x(t) = \underbrace{\cos(2t)}_{T_1} + \underbrace{\cos(3t)}_{T_2} + \underbrace{\cos(5t)}_{T_3}$

$\omega_0 = \frac{2\pi}{T}$ $\quad T_1 = \pi$ $\quad T_2 = \frac{2\pi}{3}$ $\quad T_3 = \frac{2\pi}{5}$

referans periyot alalım T_1 olsun

$\frac{T_1}{T_2} = \frac{\pi}{\frac{2\pi}{3}} = \frac{3}{2}$ $\quad \frac{T_1}{T_3} = \frac{\pi}{\frac{2\pi}{5}} = \frac{5}{2}$

$\boxed{T = n.T_1}$ $\quad T = 2T_1$

$\boxed{T = 2\pi}$

Örnek: $x[n] = \underbrace{e^{j\cdot(\frac{2\pi}{3})n}}_{N_1} + \underbrace{e^{j(\frac{3\pi}{4})n}}_{N_2}$

$x[n] = x[n+N]$

$N_1 = e^{j(\frac{2\pi}{3})\cdot(n+N)} \Rightarrow e^{j\frac{2\pi}{3}\cdot n} + e^{j\frac{2\pi}{3}\cdot N} = e^{j\frac{2\pi}{3}\cdot n}$

$e^{j2\pi k} = e^{j\frac{2\pi}{3}\cdot N}$

$2\pi k = \frac{2\pi}{3}\cdot N_1$

$N_1 = 3k$ en düşük değer yapmaya çalışıyoruz.

$N_1 = 3$

Sistemlerin Özellikleri

Memory: Girişin o anki değeri çıkışın o anki değerine eşit olursa hafızasız, eşit olmazsa hafızalı sistemdir.

Örnek: $y[\vec{n}]^{5} = 2x[\vec{n}]^{5}$ Hafızasız

$y[\vec{n}]^{15} = (2x[\vec{n}]^{15} - x^2[\vec{n}]^{15})^2$ Hafızasız

$y(\vec{t})^{10} = 2x(\vec{t}^{10} - 2)$ Hafızalı

$y[\vec{n}]^{15} = \sum_{-\infty}^{n} x[k] \Rightarrow x[-\infty] + \dots + x[0] + x[1] + \dots + x[15]$ Hafızalı

$x[15]$ olması hafızasız olacağı anlamına gelmez.

Nedensellik: Girişin o anki değeri çıkışın o anki değerine eşit veya küçük ise sistem nedenseldir.

Örnek: $\underbrace{y(t)}_{\text{çıkış}} = 2x\underbrace{(t-1)}_{\text{giriş}}$ $9 \to 10$ $-11 \to -10$ Nedensel

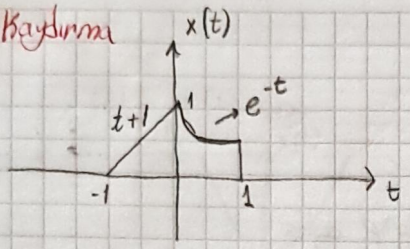
$y(t_2^2) = x(t)$ $2 \to 4$ $-3 \to 9$ $\frac{1}{4} \to \frac{1}{16}$ Nedensel değildir.

$\underbrace{y(t)}_{\text{çıkış}} = \underbrace{x(t)}_{\text{giriş}} \cdot \cos(t+1)$

$x(t+b)$

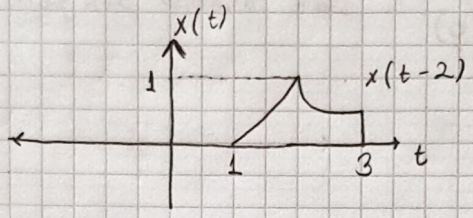
Zamanda Kaydırma

$b>0$ sola



$$x(t) = \begin{cases} t+1, & -1 \le t \le 0 \\ e^{-t}, & 0 \le t \le 1 \\ 0, & \text{diğer} \end{cases}$$

$x(t-2) = ?$



$$x(t-2) = \begin{cases} t+1-2, & -1 \le t-2 \le 0 \\ e^{-(t-2)}, & 0 \le t-2 \le 1 \\ 0, & \text{diğer} \end{cases}$$

t	$x(t)$	$x(t+3)$
1	$x(1) = 1/e$	$x(4) = 0$
0	$x(0) = 1$	$x(3) = 0$
-1	$x(-1) = 0$	$x(2) = 0$
-2	$x(-2) = 0$	$x(1) = 1/e$

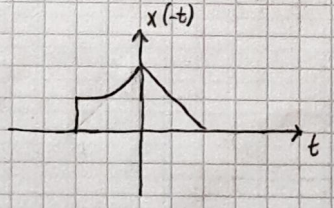
$x(t+b)$

$b>0$, b birim sola

$b<0$, b birim sağa

Zamanda Ters Çevirme

$x(t) = x(-t)$



$$x(-t) = \begin{cases} -t+1, & -1 \le -t \le 0 \\ e^{t}, & 0 \le -t \le 1 \\ 0, & \text{diğer} \end{cases}$$

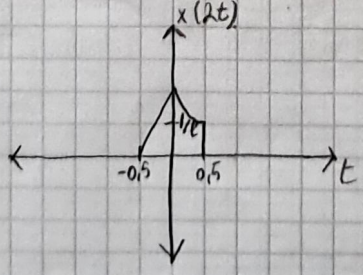
Zamanda Sıkıştırma

$x(t) = x(at)$

$a>1$, y ekseni boyunca x'e göre a çarpanı kadar sıkıştırılır.

$a<1$, y ekseni boyunca x'e göre $\frac{1}{a}$ çarpanı kadar daraltılır.

$x(t) = x(2t)$



$$x(2t) = \begin{cases} 2t+1, & -1 \le 2t \le 0 \\ e^{-2t}, & 0 \le 2t \le 1 \\ 0, & \text{diğer} \end{cases}$$

$$x(2t) = \begin{cases} 2t+1, & -1/2 \le t \le 0 \\ e^{-2t}, & 0 \le t \le 1/2 \\ 0, & \text{diğer} \end{cases}$$

Zamanda Ölçekleme ve Kaydırma

Hangisi önce yapılmalı?

$x(t) \rightarrow x(at-b)$

$x(t-b)$

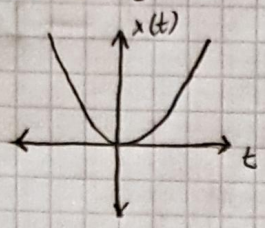
$\downarrow$

$x(at-b)$

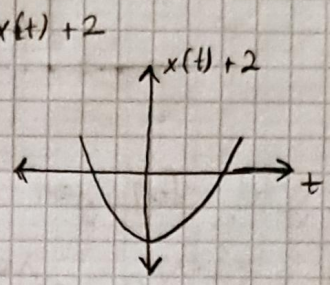
önce kaydırıp daha sonra ölçeklemek daha iyidir.

Gerlik Kaydırma

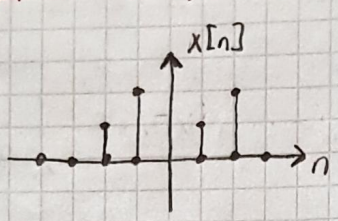
$x(t) = x(t) + b$

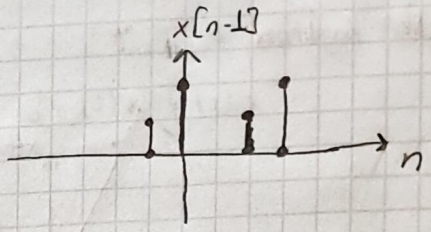


$x(t) + 2$



Ayrık Zamanlı Sinyallerin Dönüştürülmesi





Karmaşık üstel fonksiyon

$$x(t) = A \cdot e^{\lambda t}$$

$\lambda > 0$,

$\lambda < 0$,

$\lambda = 0$,

Karmaşık Sinüzoidal Fonksiyon

Euler ilişkisi

$$e^{x} = \cos x + j \sin x$$

$$e^{-x} = \cos x - j \sin x$$

$$x(t) = e^{x}$$

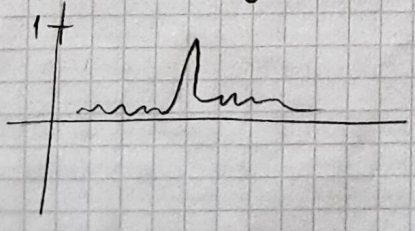
$$x_e(t) = \frac{e^{jx} + e^{-jx}}{2}$$

$$x_o(t) = \frac{e^{jx} - e^{-jx}}{2j}$$

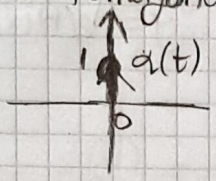
$$A \cdot e^{\lambda t} = |A| e^{j\theta} \cdot e^{(\sigma + j\omega)t}$$

$$= |A| \cdot e^{\sigma t} \cdot e^{j(\omega t + \theta)}$$

Sinüs Kardinal Fonksiyonu

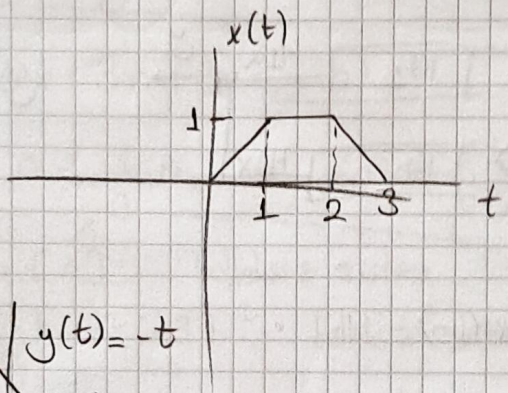


Birim Dürtü Fonksiyonu

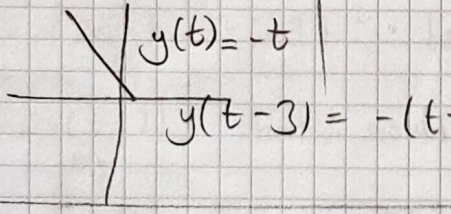


$$\int_{-\infty}^{\infty} \alpha(t)\,dt = 1$$

Delta fonksiyon özellikleri



$$x(t) = \begin{cases} t, & 0 \le x \le 1 \\ 1, & 1 \le x \le 2 \\ 3-t, & 2 \le x \le 3 \\ 0, & \text{diğer} \end{cases}$$



Ayrık Zamanlı Fonksiyonlar için:

$\frac{\pi}{\delta} = \frac{2\pi}{T}$ $\quad T = 16$

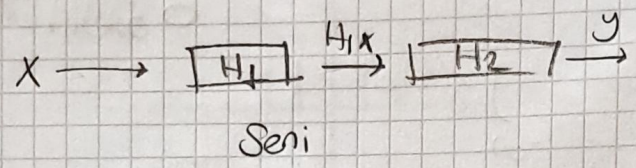
$\frac{2\pi}{1} = 0$

Özgün: $\cos(\Omega n)$ fonksiyonu CT için periyodik kabul edilmez.

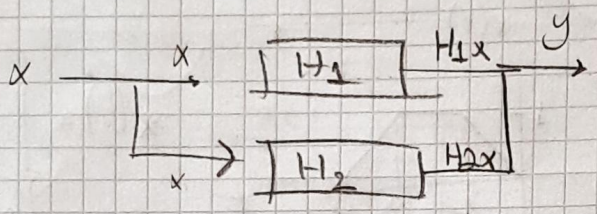
$\frac{2\pi}{7} = \frac{k\pi}{1}$

Sistemlerin Seri ve Paralel Bağlanması

$y = H_2(H_1 x)$



Seri



$y = H_1 x + H_2 x$

Belleksiz Sistem

$y(t_0) = Hx(t_0)$

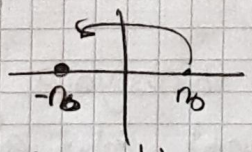
$y(n) = Hx(n) = x(-n)$

$n = n_0$ için

$y(n_0) = Hx(n_0) = x(-n_0)$ → Bellekli dir!

$Hx(n) = e^{x(n)}$

Belleksizdir.



geçmişteki bir değerin hatırlanması, belleği olduğunu gösterir.

! nedensel sistemler geçmişe bağlıdır

! anti-nedensel sistemler geleceğe bağlıdır.

$y(n) = Hx(n) = (n+1) \cdot e^{x(n-1)}$

$n = n_0$ için

$y(n_0) = Hx(n_0) = (n_0+1)\, e^{x(n_0-1)}$ nedenseldir!

6.) BIBO kararlılığı
Doğrusallık

$Hx(t) = x(t-t_0) = y(t)$

$t = t + t_0$ için

$x(t) = y(t+t_0)$ ✓ Tersine çevrilebilir bir sistem

$Hx(n) = 3x(3n+3) = y(n)$

$n = \frac{1}{3}n - 1$ dönüşümü yapılarak çözülür

$y[n] = 2x[n^{\uparrow 5}]$ belleksiz sistem

$y[n] = (2x[n] - x^2[n])^2$ belleksiz sistem

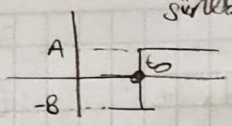
$y(t) = 2x(t-2)$ bellekli sistem

$y[n] = \sum_{k=-\infty}^{n} x[k] \Rightarrow x[-\infty] + \dots + x[0] + x[1] + \dots + x[10]$

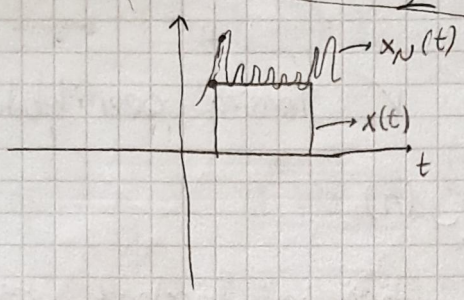
bellekli bir sistemdir.

süreksiz Fonk

$$\frac{\lim_{t_0 \to t_0^+} x(t) + \lim_{t \to t_0^-} x(t)}{2}$$



Noktasal yakınsama yapılamaz



süreksizlik noktasında
$x(t)$ ile $x_N(t)$ nasafesi değişmez.

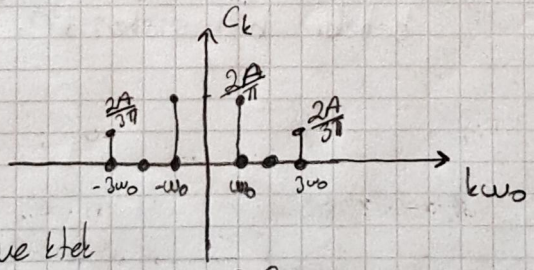
Gibbs fenomeni

Örnek: $C_k = \begin{cases} \frac{-2jA}{k\pi}, & k \text{ tek} \\ 0, & k \text{ çift} \end{cases}$

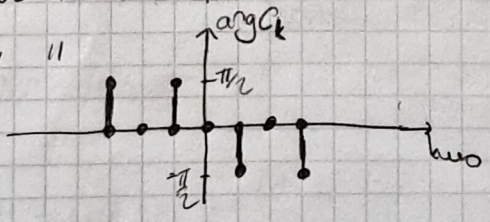
$|C_0| = 0$

$|C_k| = \begin{cases} \frac{2A}{\pi|k|}, & k \text{ tek} \\ 0, & k \text{ çift} \end{cases}$

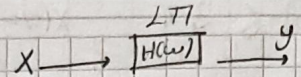
Frekans spektrumu için
$|C_k|$ $\arg C_k$



$\text{Arg } C_k = \begin{cases} \pi/2, & k<0 \text{ ve k tek} \\ -\pi/2, & k>0 \text{ " "} \\ 0, & \text{diğer} \end{cases}$



Örnek: $H(\omega) = e^{-j\omega/4}$



$x(t) = \frac{1}{2}\cos 2\pi t$

$$C_k = \begin{cases} 1/4, & k=1 \\ -1/4, & k=-1 \\ 0, & \text{diğer} \end{cases}$$

$x(t) = \frac{1}{4}e^{j2\pi t} - \frac{1}{4}e^{-j2\pi t}$

$\omega_0 = 2\pi \qquad T = 1$

$$x(t) = \sum_{k=-\infty}^{\infty} c_k \cdot e^{jk\omega_0 t}$$

$$y(t) = \sum_{k=-\infty}^{\infty} c_k H(k\omega_0)$$

$b_{-1} = c_{-1} H(-2\pi) = \frac{1}{4}e^{j2\pi/4} = -\frac{1}{4}e^{j\pi/2}$

$b_1 = c_1 H(2\pi) = \frac{1}{4}e^{-j2\pi/4} = \frac{1}{4}e^{-j\pi/2}$

$y(t) = b_{-1}e^{-j2\pi t} + b_1 e^{j2\pi t}$

$= -\frac{1}{4}e^{j\pi/2} \cdot e^{-j2\pi t} + \frac{1}{4}e^{-j\pi/2} \cdot e^{j2\pi t}$

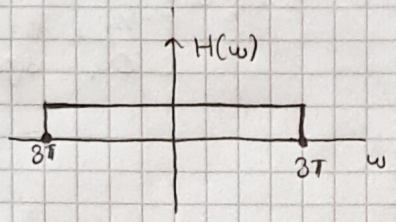
$= \frac{2j}{4}\left[\frac{e^{j(2\pi t - \pi/2)} - e^{-j(2\pi t - \pi/2)}}{2j}\right] = \frac{j}{2}\sin\left(2\pi t - \frac{\pi}{2}\right)$

Filtreleme

$$H(\omega) = \begin{cases} 1, & |\omega| \leq 3\pi \\ 0, & \text{diğer} \end{cases}$$

$x(t) = 1 + \cos 2\pi t + \cos 4\pi t + \frac{1}{2}\cos 6\pi t$

$\omega_0 = 2\pi$



$$x(t) = \underbrace{\frac{1}{k=0}}_{k=0} + 2\cdot\frac{1}{2}\left[e^{\overset{k=1}{j2\pi t}} + e^{\overset{k=-1}{-j2\pi t}}\right] + \frac{1}{2}\cdot\left[e^{\overset{k=2}{j2\cdot 2\pi t}} + e^{\overset{k=-2}{-j2\cdot 2\pi t}}\right] +$$

$$\frac{1}{2}\cdot\frac{1}{2}\left[e^{\overset{k=3}{j3\cdot 2\pi t}} + e^{\overset{k=-3}{-j3\cdot 2\pi t}}\right]$$

$$a_k = \begin{cases} 1, & k \in \{-1, 0, 1\} \\ 1/2, & k \in \{-2, 2\} \\ 1/4, & k \in \{-3, 3\} \end{cases}$$

$$x(t) = \sum_{k=-3}^{3} a_k e^{jk2\pi t}$$

$$y(t) = \sum_{k=-3}^{3} \underbrace{a_k H(k\omega_0)}_{b_k} e^{jk2\pi t}$$

$b_{-3} = a_{-3} H(-6\pi) = \frac{1}{4}\cdot 0 = 0$

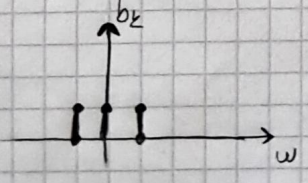
$b_3 = a_3 \cdot H(6\pi) = \frac{1}{4}\cdot 0 = 0$

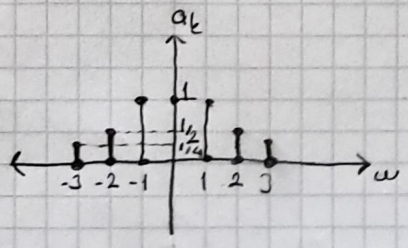
$b_2 = b_{-2} = \frac{1}{2} H(2\cdot 2\pi) = 0$

$b_1 = b_{-1} = 1 \cdot H(1\cdot 2\pi) = 1\cdot 1 = 1$

$b_0 = 1 \cdot H(0\cdot 2\pi) = 1\cdot 1 = 1$

$$b_k = \begin{cases} 1, & k \in \{-1, 0, 1\} \\ 0, & \text{diğer} \end{cases}$$





$$y(t) = \sum_{k=-1}^{1} b_k \cdot e^{jk2\pi t} = e^{j2\pi t} + 1 + e^{-j2\pi t}$$

Ayrık Zamanlı Sistemlerde Fourier Serileri

$$x(n) \Longrightarrow \sum_{k=\langle N \rangle} c_k e^{j k \frac{2\pi}{N} n} = \sum_{k=0}^{N-1} c_k e^{j k \frac{2\pi}{N} n}$$

↳ bir periyot boyunca

$$c_k = \frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\left(\frac{2\pi}{N}\right)n}$$

Örnek: $x(n) = \sin\left(\frac{2\pi}{7} n\right)$

$$\omega_0 = \frac{2\pi}{N} = \frac{2\pi}{\frac{2\pi}{7}} = 7$$

$$x(n) = \sum_{k=-3}^{3} c_k e^{jk\left(\frac{2\pi}{7}\right)n}$$

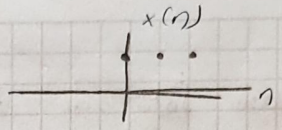
$$c_k = \frac{1}{N} \sum_{n=-3}^{3} x(n) e^{-jk\left(\frac{2\pi}{7}\right)n}$$

$$c_k = \frac{1}{7} \sum_{n=-3}^{3} \underbrace{\frac{1}{2j}\left[e^{j\frac{2\pi}{7}n} - e^{-j\frac{2\pi}{7}n} \right]}_{k = \pm 1 \text{ kısmında tanımlıdır.}} \cdot e^{-jk\frac{2\pi}{7}n}$$

$$x(n) = \underbrace{\frac{1}{2j} e^{j\frac{2\pi}{7}(1)n}}_{c_1} \underbrace{- \frac{1}{2j} e^{j\frac{2\pi}{7}(-1)n}}_{c_{-1}}$$

$$c_k = \begin{cases} -\frac{1}{2j}, & k = -1 \\ \frac{1}{2j}, & k = 1 \\ 0, & \text{diğer} \end{cases}$$

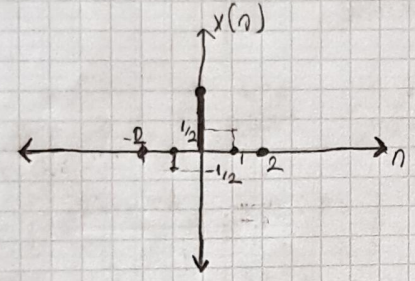
Örnek: $x(n) = \sum_{l=-\infty}^{\infty} \delta(n - N \cdot l)$



$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} \delta(n) \cdot e^{-j k \frac{2\pi}{N} n} = \frac{1}{N} \cdot e^{j \cdot 0}$$

$$c_k = \frac{1}{N}$$

Örnek:



$N = 5$

$$x(n) = \begin{cases} -1/2, & n = -1 \\ 1/2, & n = 1 \\ 1, & n = 0 \\ 0, & n \in \{-2, 2\} \end{cases}$$

$$x(n) = \sum_{k=<N} c_k \cdot e^{j k \frac{2\pi}{N} n}$$

$$c_k = \frac{1}{N} \sum_{n=-2}^{2} x(n) \cdot e^{-j k \frac{2\pi}{N} n}$$

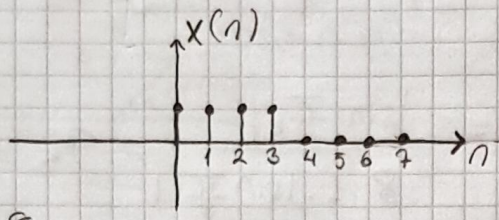
$$\frac{1}{5} \left[x(-1) \cdot e^{j k \frac{2\pi}{5}(-1)} + x(0) \cdot e^{j k \frac{2\pi}{5}(0)} + x(1) e^{-j k \frac{2\pi}{5}(1)} \right]$$

$$\frac{1}{5} \left[\frac{-1}{2} \cdot e^{j k \frac{2\pi}{5}} + 1 + \frac{1}{2} e^{-j k \frac{2\pi}{5}} \right]$$

$$\frac{1}{5} \left[1 - \frac{j}{2} \left(\frac{e^{j k \frac{2\pi}{5}} - e^{-j k \frac{2\pi}{5}}}{2j} \right) \right]$$

$$= \frac{1}{5} \left[1 - j \left(\sin \frac{2\pi}{5} k \right) \right]$$

Örnek: $x(n) = \begin{cases} 1, & n \in [0 \dots 3] \\ 0, & n \in [4 \dots 7] \end{cases}$



$N = 8$

$$c_k = \frac{1}{N} \sum_{n=0}^{7} x(n) \cdot e^{-jk \frac{2\pi}{N} \cdot n}$$

$$c_k = \frac{1}{8} \sum_{n=0}^{3} x(n) \cdot e^{-jk \frac{2\pi}{8} \cdot n} = \frac{1}{8} \sum_{n=0}^{3} 1 \cdot e^{-j \frac{2\pi}{8} kn} =$$

$$c_k = \frac{1}{8} \left[1 \cdot e^{-jk \frac{\pi}{4} \cdot (0)} + 1 \cdot e^{-jk \frac{\pi}{4} \cdot 1} + 1 \cdot e^{-jk \frac{\pi}{4} \cdot 2} + 1 \cdot e^{-jk \frac{\pi}{4} 3} \right]$$

$$\frac{1}{8} \left[1 + e^{-jk \frac{\pi}{4}} + e^{-jk \frac{\pi}{2}} + e^{-jk \frac{3\pi}{4}} \right]$$

$$\frac{1}{8} \sum_{n=0}^{3} e^{jk \frac{\pi}{4}} \Rightarrow \frac{1}{8} \cdot \left[\frac{1 - e^{-jk\pi}}{1 - e^{-jk\pi/4}} \right]$$

$$c_k = \frac{1}{8} \cdot \frac{e^{-jk\frac{\pi}{2}} \left(e^{jk\frac{\pi}{2}} - e^{-jk\pi/2} \right)}{e^{-jk\pi/8} \left(e^{jk(\pi/8)} - e^{-jk\pi/8} \right)} = \frac{1}{8} \frac{\sin\left(\frac{\pi}{2}k\right)}{e^{jk\frac{3\pi}{8}} \sin\left(\frac{\pi}{8}k\right)}$$

$$c_0 = \frac{1}{N}$$

$$c_k = \begin{cases} 1/2, & k = 0 \\ \frac{1/8 \; 1 - e^{-jk\pi}}{1 - e^{-jk\pi/4}}, & k \neq 0 \end{cases}$$

Örnek: $a_k = \frac{1}{8e^{-j\frac{3\pi}{8}k} \cdot \sin(k\frac{\pi}{8})} \cdot \sin(k\pi/2)$

Frekans spektrumu nedir?
Faz spektrumu nedir?

$k.\Omega_0 = k.\frac{2\pi}{8} = k\frac{\pi}{4}$

$$|a_k| = \left| \frac{\sin(k\pi/2)}{8e^{-j\frac{3\pi}{8}k} \cdot \sin(\frac{\pi}{8}k)} \right|$$

$$= \frac{|\sin(\frac{\pi}{2}k)|}{|8| \cdot |e^{-j\frac{3\pi}{8}k}| \cdot |\sin(\frac{\pi}{8}k)|} = \frac{|\sin\frac{\pi}{2}k|}{8 \cdot 1 \cdot |\sin\frac{\pi}{8}k|} = a_k$$

$|a_0| = 1/2$

$$|a_k| = \begin{cases} 1/2, & k=0 \\ \frac{\sin \pi/2 k}{8|\sin\frac{\pi}{8}k|}, & k \neq 0 \end{cases}$$

$$\arg(\sin\tfrac{\pi}{2}k) - \arg(8.e^{-j\frac{3\pi}{8}k}) - \arg(|\sin\tfrac{\pi}{8}k)$$

$$\frac{3\pi}{8}k$$

Örnek: $H(-\Omega) = e^{-j-\Omega}$

$x(n) = \frac{1}{2}\cos\frac{2\pi}{5}n$ girişi, yukarıdaki sisteme uygulanırsa çıkış ne olur?

$$x(n) = \frac{1}{2}\cdot\frac{1}{2}\left(e^{j\frac{2\pi}{5}n} + e^{-j\frac{2\pi}{5}n}\right) \qquad N=5$$

$$a_k = \begin{cases} 1/4, & k \in \{-1, 1\} \\ 0, & k \in \{-2, 0, 2\} \end{cases}$$

$$y(n) = \sum_{k=\langle N \rangle} \underbrace{a_k \cdot H(k\Omega_0)}_{b_k} e^{jk\Omega_0 n}$$

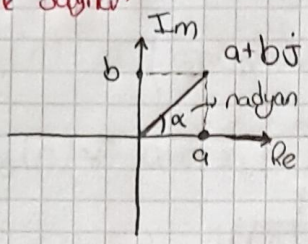
$b_{-1} = a_{-1} \cdot H(-\frac{2\pi}{5}) = \frac{1}{4}e^{j\frac{2\pi}{5}}$

$b_1 = a_1 \cdot H(\frac{2\pi}{5}) = 1/4\, e^{-j\frac{2\pi}{5}}$

$b_{-2} = b_2 = b_0 = 0$

Udemy için Sinyaller

Karmaşık Sayılar



$a+bj = r \cdot e^{j\alpha}$

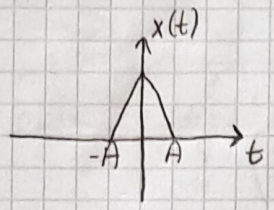
$r = \sqrt{a^2+b^2}$

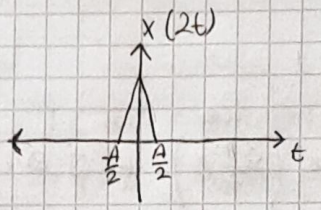
$\alpha = \arctan\left(\frac{b}{a}\right)$

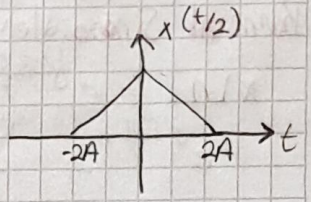
$a = r \cdot \cos\alpha$

$b = r \cdot \sin\alpha$

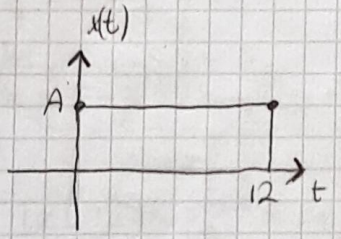
Ölçekleme





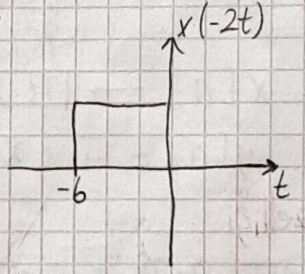


Örnek:

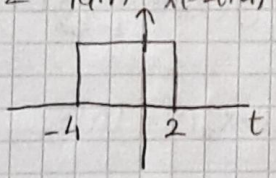


$x(-2t+4) = ?$

$x = -2t$



$t = t-2$ için $x(-2t+4)$



$-2t+4 = 12$

$t = -4$

$-2t+4 = 0$

$t = 2$

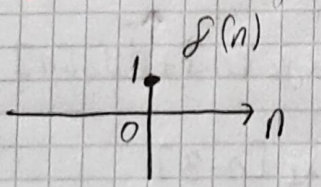
$T = 25$

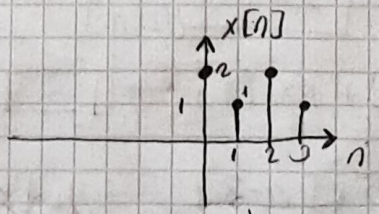
$\frac{2\pi}{\omega_0} = 25$

$25\omega_0 = 2\pi$

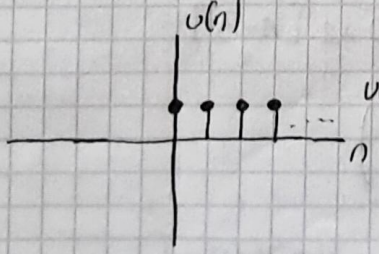
$\frac{2\pi}{25}$

Dürtü Sinyali





$x[n] = 2\cdot\delta(n) + \delta(n-1) + 2\cdot\delta(n-2) + \delta(n-3)$



$u(n) = \sum_{k=0}^{\infty} \delta(n-k)$

↳ dürtü cinsinden yazılması

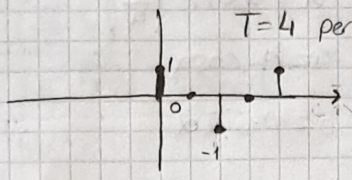
$\delta(n) = u(n) - u(n-1)$

Ayrık Zamanlı Sinüzoidal Sinyaller

$F_s = 200\ Hz$

$x(t) = \cos(100\pi t)$

$x[n] = \cos\left(100\pi \cdot \frac{1}{200} \cdot n\right) = \cos\left(\frac{\pi}{2}n\right)$ $\left. x(t) \right|_{t = n \cdot T_s}$



Karmaşık Sinüzoidler

$x[n] = e^{j\frac{3\pi}{7}n}$

$e^{j\frac{3\pi}{7}n} = \underbrace{\cos\left(\frac{3\pi}{7}n\right)}_{Re} + j\underbrace{\sin\left(\frac{3\pi}{7}n\right)}_{Im}$

Sistemler

$x[n] \longrightarrow \boxed{T\{\cdot\}} \longrightarrow y[n]$

$y[n] = T\{x[n]\}$

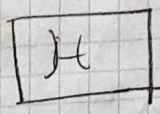
Hafıza Özelliği

$y[n] = x[n]^2$ → hafızasız

$y[n] = x[n-1]$ → hafızalı

$y[n] = \sum_{k=-\infty}^{n} x[k]$ → hafızalı

Doğrusallık

$x_1[n] \longrightarrow$  $\longrightarrow y_1[n]$

$x_2[n] \longrightarrow$ $\longrightarrow y_2[n]$

$\alpha x_1[n] + \beta x_2[n] \longrightarrow \boxed{} \longrightarrow \alpha y_1[n] + \beta y_2[n]$

"Örnek: $y[n] = 5x[n-3]$

$y_1[n] = 5x_1[n-3]$

$y_2[n] = 5x_2[n-3]$

$x[n] = \alpha x_1[n] + \beta x_2[n] \rightarrow y_3[n]$

$y_3[n] = 5 \cdot \left(\alpha \underbrace{x_1[n-3]}_{y_1[n]} + \beta \underbrace{x_2[n-3]}_{y_2[n]} \right)$

$y_3[n] = y_1[n] \cdot \alpha + \beta y_2[n]$ ✓ doğrusal

Örnek: $y(n) = a \cdot x(n) + b$

$y_1(n) = a \cdot x_1(n) + b$

$y_2(n) = a \cdot x_2(n) + b$

$y_3(n) = a \cdot \left(\alpha \cdot x_1(n) + \beta \cdot x_2(n) \right) + b$

$y_3(n) = \underline{a \cdot \alpha \, x_1(n)} + \underline{a \cdot \beta \cdot x_2(n)} + \underline{b}$

$y_2(n) + b$

$y_3(n) = y_1(n) + y_2(n) + b$ X doğrusal değil

Nedensellik

$y(t) = 2x\left(\frac{t-1}{9}\right)$ $\quad {}^{10}_{-11}$ $\quad {}^{10}_{-10}$

$9 < 10$ nedensel bir sistem

$-11 < -10$ nedensel

$y(t^2) = 2x(t)$				
9	3	$4 < 16$	nedensel	$\frac{1}{4} < \frac{1}{16}$ X nedensel değil
16	4	$3 < 9$	nedensel	
1/16	1/4	$1 \leq 1$	nedensel	

Kararlılık

$|x[n]| < M$ ise kararlıdır.

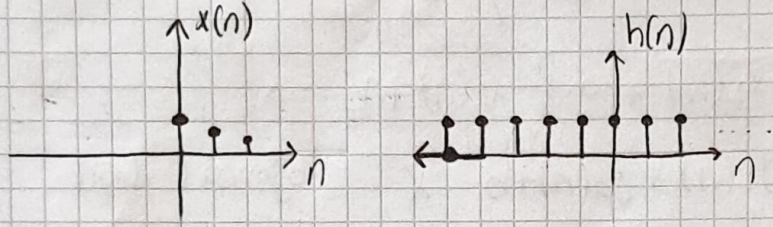
$y[n] = 100 \underbrace{x[n]}_{k} + 50 \underbrace{x[n-1]}_{k}$ $\quad y[n] < 150k$ kararlı

$y(n) = \sum_{n=-\infty}^{\infty} n \cdot x(n)$ $\quad |x(n)| < L$ kararsız

Konvolüsyon DT

$$y(n) = \sum x(k)\, h(n-k) = \sum h(k)\, x(n-k)$$

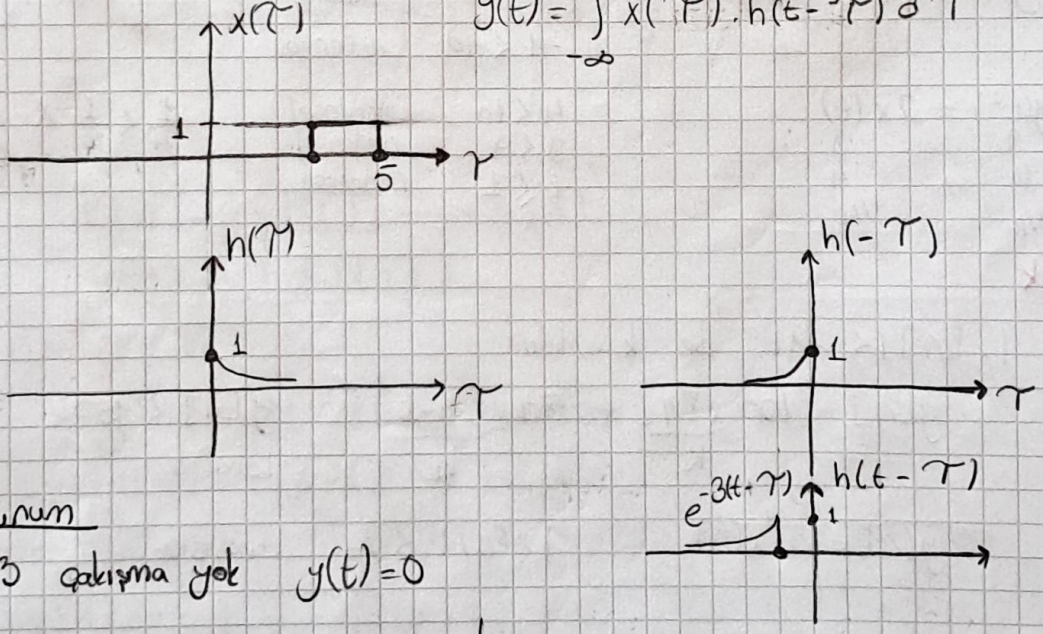
Örnek: $x(n) = \left(\frac{1}{2}\right)^n u(n)$, $h(n) = u(-n+5)$ $y(n) = ?$



Örnek: $x(t) = u(t-3) - u(t-5)$ $h(t) = e^{-3t}\, u(t)$

a) $y(t) = x(t) * h(t)$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

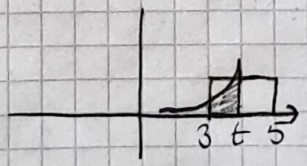


1. durum

$t < 3$ çakışma yok $y(t) = 0$

2 durum

$3 < t < 5$



$$\int_3^t 1 \cdot e^{-3(t-\tau)}\, d\tau$$

$$\int_3^t e^{-3t} \cdot e^{3\tau}\, d\tau = e^{-3t} \cdot \frac{e^{3\tau}}{3} \Big|_3^t = e^{-3t}\left(\frac{e^{3t}}{3} - \frac{e^{9}}{3}\right)$$

$$y(t) \quad 3<t<5 \quad = \frac{1 - e^{-3t+9}}{3}$$

3. durum

$t > 5$

Sinyallerde Enerji ve Güç

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt \qquad P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 \, dt$$

$$E = \sum_{n=-\infty}^{\infty} |x[n]|^2 \qquad P = \lim_{N\to\infty} \sum_{-N}^{N} |x[n]|^2$$

Örnek: $x(t) = e^{-2t} \cdot u(t)$

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, dt \Rightarrow \int_{-\infty}^{+\infty} |e^{-2t} u(t)|^2 \, dt = \int_{0}^{\infty} e^{-4t} \, dt$$

$$= \frac{-1}{4} e^{-4t} \Big|_0^{\infty}$$

$$\frac{1}{\infty} - \left(\frac{-1}{4}\right) = \frac{1}{4}$$

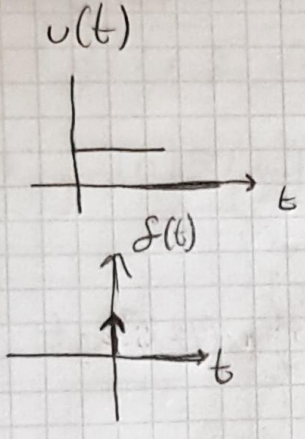
$$P = \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |x(t)|^2 \, dt$$

$$\Rightarrow \lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} |e^{-2t} u(t)|^2 \, dt$$

$$\lim_{T\to\infty} \frac{1}{2T} \int_{-T}^{T} e^{-4t} \, dt = 0$$

Örnek: $x(t) = \cos(t)$

$$E = \int_{-\infty}^{\infty} |\cos(t)|^2 \, dt = \int_{-\infty}^{\infty} \cos^2(t) \, dt = \infty$$



$$u(t) = \begin{cases} 1, & t>0 \\ 0, & t<0 \end{cases}$$

$$\frac{d\,u(t)}{dt}$$

Nedensellik

$y(t) = 3x(t) - 0.5x(t-5)$ nedensel

$y(t) = 3x(t+2)$ → gelecekteki değer nedensel değil

$y[n] = \cos(\pi(n+1))\,x[n]$ nedensel

Kararlılık

$|x(t)| < M$ diyebiliyorsak sistem kararlıdır.

Doğrusallık

Örnek: $y(t) = 2x(t-1)$

$y_1(t) = 2x_1(t-1)$

$y_2(t) = 2x_2(t-1)$

$x_3(t) = \alpha \cdot x_1(t) + \beta \cdot x_2(t)$

$y_3(t) = 2x_3(t-1)$

$\Rightarrow y_3(t) = 2(\alpha x_1(t-1) + \beta \cdot x_2(t-1))$

$y_3(t) = \underbrace{2\alpha \cdot x_1(t-1)}_{y_1(t)} + \underbrace{2\beta x_2(t-1)}_{y_2(t)}$

doğrusal

Zamanda Değişmezlik

Örnek: $y(t) = x(2t)$

$y_2(t) = x_2(2t) = x(2t - t_0)$

$x_2(t) = x(t - t_0)$

$y(t - t_0) = x(2t - 2t_0)$

eşit değil → zamanda değişen sistem

Örnek: $y(t) = x^2(t)$

$y_2(t) = x_2^2(t) = x_2^2(t - t_0)$

$x_2(t) = x(t - t_0)$

$y(t - t_0) = x^2(t - t_0)$

eşit → zamanda değişmez sistem

Hafızasızlık

$y[n] = x^2[n]$ hafızasız

$y[n] = x[n]\cos(n - 5)$ hafızasız

$y(n) = x(n - 1)$ hafızalı

$y(t) = \cos(3t) \cdot x(t)$ hafızasız ✓

$y_2(t) = \cos(3t) \cdot x_2(t) = \cos(3t - t_0) \cdot x_2(t - t_0)$

$x_2(t) = x(t - t_0)$ → zamanda değişen ✓

$y(t - t_0) = \cos(3t - 3t_0) \cdot x(t - t_0)$

$y_1(t) = \cos 3t \cdot x_1(t)$

$y_2(t) = \cos 3t \cdot x_2(t)$

$x_3(t) = \alpha \cdot x_1(t) + \beta \cdot x_2(t)$

$y_3(t) = \cos 3t \cdot x_3(t)$

$\cos 3t\,(\alpha \cdot x_1(t) + \beta \cdot x_2(t))$

$\cos 3t\, \alpha \cdot x_1(t) + \beta \cdot \cos 3t \cdot x_2(t)$

$y_1(t)$ $y_2(t)$

nedensel ✓

kararlıdır. ✓

doğrusal ✓

Konv Örnek: $x[n] = \left(\frac{1}{2}\right)^{n-2} \cdot u[n-2]$

$h[n] = u[n+2]$ $\qquad x[n] * h[n]$

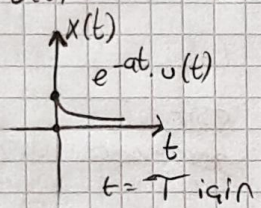
$$y[n] = \sum_{k=-\infty}^{\infty} h[k]\, x[n-k]$$

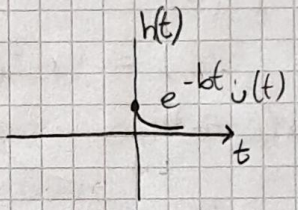
$h[k] = u[k+2]$

Örnek: $x(t) = e^{-at} \cdot u(t)$

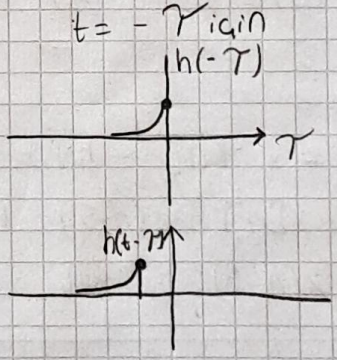
$h(t) = e^{-bt} \cdot u(t)$

$$\int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, d\tau$$





$$y(t) = \int_{-\infty}^{\infty} e^{-a\tau} u(\tau) \cdot e^{-b(t-\tau)}\, d\tau$$



$\tau > 0$

$t - \tau > 0$

$\tau < t$ $\qquad t<0$ ise $\tau \ll 0$

Durum 1: $t < 0$ ise

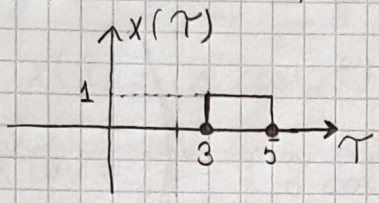
$y(t) = 0$

Durum 2: $t > 0$ ise

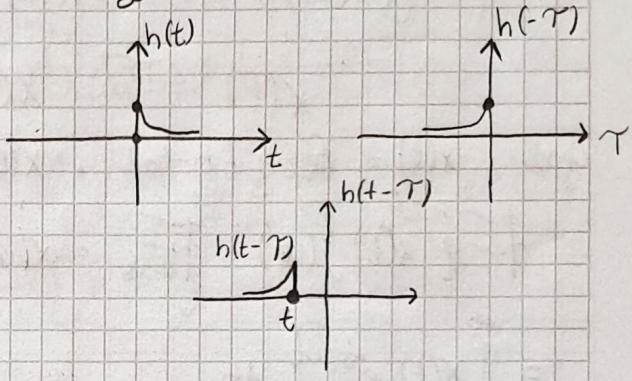
$$\int_0^t e^{-a\tau} \cdot e^{-b(t-\tau)}\, d\tau \Rightarrow +e^{-a} \cdot \frac{e^{\tau}}{\tau} \cdot +e^{-bt} \cdot e^{b} \cdot \frac{e^{\tau}}{\tau} \Big|_0^{\epsilon}$$

Örnek: $x(t) = u(t-3) - u(t-5)$

$h(t) = e^{-3t} \cdot u(t)$



$$\int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, d\tau$$



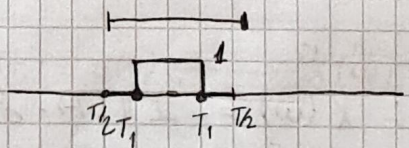
Fourier Serileri

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \cdot e^{j\omega_0 k t}$$

$$a_k \frac{1}{T} \int_T x(t) \cdot e^{-jk\omega_0 t}\, dt$$

Örnek:

$$x(t) = \begin{cases} 1, & |t| < T_1 \\ 0, & T_1 < |t| < \frac{T}{2} \end{cases}$$



$$a_k = \frac{1}{T} \int_T x(t) \cdot e^{-jk\omega_0 t}\, dt \longrightarrow$$

$$a_k = \frac{1}{T} \int_{-T_1}^{T_1} 1 \cdot e^{jk\omega_0 t}\, dt$$

$$= \frac{-1}{Tjk\omega_0} \cdot e^{-j\omega_0 k t} \Big|_{-T_1}^{T_1}$$

$$\frac{-1}{Tjk\omega_0} \left(e^{-jk\omega_0 T_1} - e^{jk\omega_0 T_1} \right)$$

$$\frac{2j}{Tjk\omega_0} \left(\frac{e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}}{2j} \right)$$

$$= \frac{2}{T k\omega_0} \cdot \sin(\omega_0 k T_1)$$

$\frac{2\pi}{\omega_0}$

$T = \frac{2\pi}{\omega_0}$ için

$$\frac{1}{\pi k} \cdot \sin(\omega_0 k T_1)$$

sinc

$k = 1$ için

$2 \cdot \cos(\omega_1 t$

$T = \frac{2\pi}{\omega_0}$

$\omega_0 = \frac{2\pi}{8}$

$\omega_0 = \frac{\pi}{4}$

CTFT dönüşümü

$\mathcal{F}\{x(t)\}(\omega) = X(\omega)$ gösterimi

$$x(t) \xrightarrow{CTFT} X(\omega)$$

Örnek: $x(t) = A\,\delta(t-t_0)$, $X(\omega) = ?$

$$\mathcal{F}\{x(t)\}(\omega) = \mathcal{F}_x = X(\omega)$$

$$= \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$\Rightarrow \int_{-\infty}^{\infty} A\,\delta(t-t_0)\, e^{-j\omega t}\, dt = A.e^{-j\omega t}\Big|_{t=t_0}$$

$$X(\omega) = A.e^{-j\omega t_0}$$

$t_0 = 1$ için $X(\omega) = A.e^{-j\omega}$

Örnek: $X(\omega) = 2\pi A.\delta(\omega-\omega_0)$ $x(t) = ?$

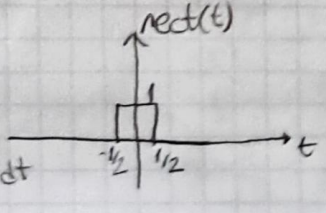
$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\omega)\, e^{j\omega t}\, d\omega$$

$$\Rightarrow \frac{1}{2\pi}\int 2\pi A.\delta(\omega-\omega_0)\, e^{j\omega t}\, d\omega$$

$$= A.e^{j\omega t}\Big|_{\omega=\omega_0} = A.e^{j\omega t} \rightarrow \omega_0 = 1 \text{ için } x(t) = A.e^{jt}$$

Örnek: $x(t) = rect(x)$ $X(\omega) = ?$

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt \Rightarrow \int_{-\infty}^{\infty} rect(t).e^{-j\omega t}\, dt$$



CTFT Özellikleri

$e^{j\omega_0 t} \xrightarrow{CTFT} 2\pi\,\delta(\omega-\omega_0)$

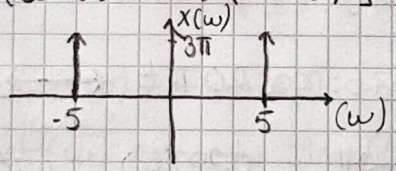
Örnek: $x(t) = 3.\cos(5t) \longleftrightarrow X(\omega) = ?$

$$X(\omega) = \mathcal{F}\{3.\cos(5(t))\} = \mathcal{F}\left\{\frac{3}{2}e^{j5t} + \frac{3}{2}e^{-j5t}\right\}$$

$$= \frac{3}{2}\mathcal{F}\{e^{j5t}\} + \frac{3}{2}\mathcal{F}\{e^{-j5t}\}$$

$$X(\omega) = 2\pi\cdot\frac{3}{2}\delta(\omega-5) + 2\pi\cdot\frac{3}{2}\delta(\omega+5)$$

$$= 3\pi[\delta(\omega-5) + \delta(\omega+5)]$$



Örnek: $1 \xrightarrow{CTFT} 2\pi\,\delta(\omega)$ $\qquad x(t) = u(t) \longleftrightarrow X(\omega) = ?$

$sgn(t) \xrightarrow{CTFT} \frac{2}{j\omega}$

$$\mathcal{F}_x = \mathcal{F}\left\{\frac{1}{2} + \frac{1}{2}sgn(t)\right\}$$

$$= \frac{1}{2}\mathcal{F}\{1\} + \frac{1}{2}\mathcal{F}\{sgn(t)\}$$

$$\frac{1}{2}\cdot 2\pi\,\delta(\omega) + \frac{1}{2}\cdot\frac{2}{j\omega}$$

$$= \pi\,\delta(\omega) + \frac{1}{j\omega} = X(\omega)$$

Zamanda Kaydırma Özelliği

Örnek: $x(t) = A \cdot \cos(\omega_0 t + \theta) \longleftrightarrow X(\omega)$

$= A \cdot \cos\left(\omega_0 \left(t + \frac{\theta}{\omega_0}\right)\right)$

$v(t) = \cos(\omega_0 t) \longleftrightarrow V(\omega) = \pi \cdot [\delta(\omega-\omega_0) + \delta(\omega+\omega_0)$

$x(t) = A \cdot v\left(t + \frac{\theta}{\omega_0}\right) \longleftrightarrow A \cdot e^{j\frac{\theta}{\omega_0}\omega} \pi \left[\delta(\omega-\omega_0) + \delta(\omega+\omega\right.$

$A = 2. \quad \omega_0 = 3\pi \quad \theta = \frac{\pi}{4}$

$X(\omega) = 2 \cdot e^{j\frac{\pi}{12}\cdot\omega} \cdot \pi \left[\delta(\omega-3\pi) + \delta(\omega+3\pi)\right]$

Frekansta Kaydırma Özelliği

Örnek: $x(t) = \cos\omega_0 t \cdot \cos 20\pi t \longleftrightarrow X(\omega)$

$x(t) = \frac{1}{2}\left(e^{j20\pi} + e^{-j20\pi t}\right) \cdot \cos(\omega_0 t)$

$\frac{1}{2} \cdot \cos\omega_0 t \cdot e^{j20\pi t} + \frac{1}{2} \cos\omega_0 t\, e^{-j20\pi t}$

$\mathcal{F}\{\cos\omega_0 t\} = \underbrace{\pi \cdot [\delta(\omega-\omega_0) + \delta(\omega+\omega_0]}_{X_1(\omega)}$

$\mathcal{F}\{e^{j20\pi t} \cdot \cos\omega_0 t\} = X_1(\omega - 20\pi)$

$X_1(\omega-20\pi) = \pi \cdot [\delta(\omega - 20\pi - \omega_0) + \delta(\omega+\omega_0-20\pi)$

$X(\omega) = \frac{1}{2\pi}\pi \left[\delta(\omega-\omega_0-20\pi) + \delta(\omega+\omega_0-20\pi) + \right.$
$\left.\delta(\omega-\omega_0+20\pi) + \delta(\omega+\omega_0+20\pi.\right]$

Ölçekleme Özelliği

Örnek: $rect(t) \xleftrightarrow{CTFT} sinc\left(\frac{\omega}{2}\right)$

$x(t) = rect(at) \longleftrightarrow X(\omega)$ $\quad a = 5$

$v(t) = rect(t) \longleftrightarrow X(\omega) = sinc\left(\frac{\omega}{2}\right)$

$x = v(at) \longleftrightarrow \frac{1}{|a|} V\left(\frac{\omega}{a}\right) = \frac{1}{5} sinc\left(\frac{\omega}{2.5}\right)$

Dualite Özelliği

Örnek: $rect(t) \xleftrightarrow{CTFT} sinc\left(\frac{\omega}{2}\right)$

$2\pi\, rect(-\omega) \longleftrightarrow sinc\left(\frac{t}{2}\right)$

Çift fonk özelliği $\leftarrow 2\pi\, rect(\omega) \longleftrightarrow sinc\left(\frac{t}{2}\right)$

Konvolüsyon Özelliği

Örnek: $x(t) = e^{-2t} \cdot u(t)$

$h(t) = u(t)$

$y(t) = h(t) * x(t)$

$y(\omega) = X(\omega) \cdot H(\omega)$

$X(\omega) = \frac{1}{2+j\omega}$, $H(\omega) = \frac{1}{j\omega} + \pi \cdot \delta(\omega)$

$$X(\omega) \cdot H(\omega) = \frac{1}{2+j\omega}\left(\frac{1}{j\omega} + \pi \cdot \delta(\omega)\right)$$

$$= \frac{1}{-\omega^2 + 2j\omega} + \frac{\pi \cdot \delta(\omega)}{2} = \frac{1}{2+j\omega}$$

$\delta(\omega) = \frac{1}{2}\delta(\omega)$

$\omega = 0$ için

Zamanda Çarpma Özelliği $x_1(t) \cdot x_2(t) \xleftarrow{CTFT} \frac{1}{2\pi} X_1 * X_2(\omega)$

Örnek: $y(t) = x(t) \cdot \cos \omega_c(t)$, $Y(\omega) = ?$

$v(t) = \cos \omega_c t$

$V(\omega) = \pi \cdot [\delta(\omega - \omega_c) + \delta(\omega + \omega_c)]$

$y(t) = x(t) \cdot v(t)$

$Y(\omega) = \frac{1}{2\pi} X(\omega) * V(\omega)$

$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\tau)\, V(\omega - \tau)\, d\tau$

$Y(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\tau)\, \pi [\delta(\omega - \tau - \omega_c) + \delta(\omega - \tau + \omega_c)]\, d\tau$

$= \frac{1}{2} \int_{-\infty}^{\infty} X(\tau) \cdot [\delta(\tau - (\omega - \omega_c)) + \delta(\tau - (\omega + \omega_c))]\, d\tau$

$= \frac{1}{2} \left[X(\tau)\Big|_{\tau = \omega - \omega_c} + X(\tau)\Big|_{\tau = \omega + \omega_c} \right]$

$= \frac{1}{2} X(\omega - \omega_c) + \frac{1}{2} X(\omega + \omega_c)$

Zamanda Türev Alma

Örnek: $x(t) = \frac{d f(t)}{dt}$, $X(\omega)$

$v(t) = f(t) \longleftrightarrow V(\omega) = 1$

$x(t) = \frac{d v(t)}{dt} \longleftrightarrow X(\omega) = j\omega V(\omega)$

$= j\omega \cdot 1 = j\omega$

Zamanda İntegral Alma

Örnek: $x(t) = v(t)$

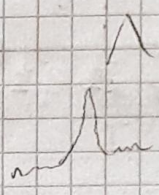
$$x(t) = v(t) = \int_{-\infty}^{t} \delta(\tau)\, d\tau$$

$$v(t) = u(t)$$

$$V(\omega) = \frac{1}{j\omega} + \pi \cdot V(0) \cdot \delta(\omega)$$

$$X(\omega) = \frac{1}{j\omega} V(\omega) + \pi \cdot V(0) \cdot \delta(\omega)$$

$$= \frac{1}{j\omega} + \pi\, \delta(\omega)$$



Frekansta Türev Alma

Örnek: $x(t) = t \cdot \cos(\omega_0 t) \longleftrightarrow X(\omega) = ?$

$$v(t) = \cos(\omega_0 t) \longleftrightarrow V(\omega) = \pi \cdot [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$x(t) = t \cdot v(t) \longleftrightarrow X(\omega) = j \frac{d}{d\omega} \left[\pi \cdot \delta(\omega - \omega_0) + \pi\, \delta(\omega + \omega_0)\right]$$

Parseval İlişkisi

Örnek: $x(t) = \text{sinc}\left(\frac{t}{2}\right)$

$$E = \int_{-\infty}^{\infty} \left|\text{sinc}\frac{t}{2}\right|^2 dt$$

$$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\omega)|^2\, d\omega$$

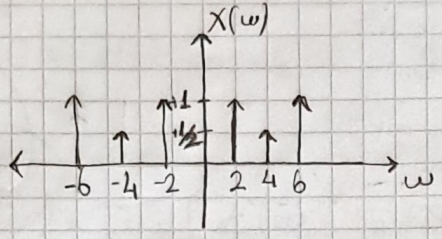
$$\text{sinc}\frac{t}{2} \xleftarrow{\text{CTFT}} X(\omega) = 2\pi\, \text{rect}(\omega)$$

$$E = \frac{1}{2\pi} \int_{-\infty}^{\infty} |2\pi\, \text{rect}(\omega)|^2\, d\omega = \frac{1}{2\pi} 4\pi^2 \int_{-1/2}^{1/2} \text{rect}^2(\omega)\, d\omega$$

$$= 2\pi \int_{-1/2}^{1/2} d\omega = 2\pi$$

Periyodik Fonksiyonların Fourier Dönüşümü

Örnek



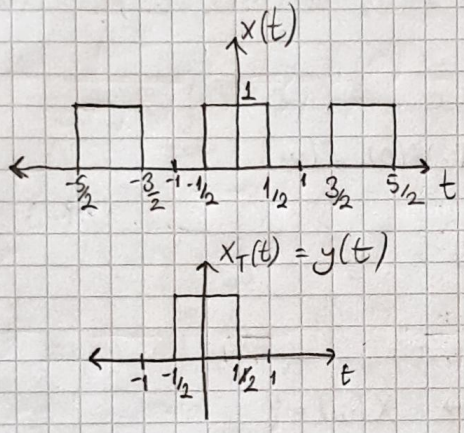
$x(t)$ periyodik ise

$$X(\omega) = \sum_{k=-\infty}^{\infty} a_k \, \delta(\omega - k\omega_0)$$

şeklinde olmalıdır.

$\omega_0 = 2$ $\quad T = \frac{2\pi}{2} = \pi$

Örnek:



$X(\omega) = ?$

$y(t) = \text{rect}(t) \longleftrightarrow \text{sinc}\frac{\omega}{2} = Y(\omega)$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T} Y(k\omega_0) = \frac{1}{2} Y(k\pi) = \frac{1}{2} \text{sinc}\left(\frac{k\pi}{2}\right)$$

$$X(\omega) = \sum_{k=-\infty}^{\infty} 2\pi a_k \, \delta(\omega - k\pi) = \sum_{k=-\infty}^{\infty} 2\pi \frac{1}{2} \sin\left(\frac{k\pi}{2}\right) \delta(\cdots)$$

Örnek: $x(t) \longleftrightarrow X(\omega)$, $y(t) \longleftrightarrow Y(\omega)$

$y(t) = \frac{d^2 x(t-2)}{dt^2}$ $\qquad Y(\omega) = ?$

$x(t) \longleftrightarrow X(\omega)$

$x(t-2) \longleftrightarrow e^{-j2\omega} \cdot X(\omega)$

$$\frac{d^2 x(t-2)}{dt^2} \longleftrightarrow (j\omega)^2 e^{-j2\omega} X(\omega) = -\omega^2 e^{-j2\omega} \cdot X(\omega)$$

Örnek: $y(t) = x(at - b)$, $y(\omega)$

$v_1(t) = x(t - b)$

$y(t) = v_1(at)$ $\qquad x(t) \longleftrightarrow X(\omega)$

$v_1(t) = x(t-b) \longleftrightarrow e^{-jb\omega} \cdot X(\omega) = V_1(\omega)$

$y(t) = v_1(at) \longleftrightarrow \frac{1}{|a|} V_1\left(\frac{\omega}{a}\right) = y(\omega)$

$y(\omega) = \frac{1}{|a|} e^{j\frac{b}{a}\omega} \cdot X\left(\frac{\omega}{a}\right)$

Örnek: $X(t) = \sum_{k=-\infty}^{\infty} x_0(t - kT)$, $x_0(t) = A\,\text{rect}\left(\frac{2t}{T}\right)$

$X(\omega) = \mathcal{F}\left\{ \sum_{k=-\infty}^{\infty} x_0(t - kT) \right.$

$=$

$x_0(t) = A \cdot \text{rect}\left(\frac{2t}{T}\right) \longleftrightarrow \frac{A \cdot T}{2} \text{sinc}\left(\frac{T\omega}{4}\right) = X_0(\omega)$

$\text{rect}(t) \longleftrightarrow \text{sinc}\left(\frac{\omega}{2}\right)$

$\text{rect}(at) \longleftrightarrow \frac{1}{a} \text{sinc}\frac{\omega}{2a}$

$a = \frac{2}{T}$ $\text{rect}\frac{2t}{T} \longleftrightarrow \frac{T}{2} \text{sinc}\frac{T\omega}{4}$

Örnek: $x(t) = \int_{-\infty}^{t} e^{-(3+j2)\tau} u(\tau)\, d\tau$

$u_1(t) = e^{-(3+j2)t} \cdot u(t)$

$u_1(t) = e^{-3t} \cdot e^{-j2t} \cdot u(t)$

$\frac{1}{3+j\omega} = u_2(\omega)$

$u_1(\omega) = u_2(\omega+2)$

$= \frac{1}{3+j(\omega+2)} \Rightarrow U_1(0) = \frac{1}{3+j2}$

$x(t) = \int_{-\infty}^{t} u_1(\tau)\, d\tau \longleftrightarrow X(\omega) = \frac{U_1(\omega)}{j\omega} + \pi \cdot U_1(0) \cdot \delta(\omega)$

$X(\omega) = \frac{1}{j\omega} \cdot \frac{1}{3+j(\omega+2)} + \pi \cdot \frac{1}{3+j2} \delta(\omega)$

FT ve Frekans spektrumu

Örnek: $x(t) = \text{sinc}\left(\frac{\omega}{2}\right) \longleftrightarrow X(\omega) = 2\pi \cdot \text{rect}(\omega)$

Örnek: $x(t) = \frac{1}{2}\,\text{sgn}(t-1)$ $\qquad \text{sgn}\, t \overset{CTFT}{\longleftrightarrow} \frac{2}{j\omega}$

$\text{sgn}(t-1) \longleftrightarrow e^{-j\omega} \cdot \frac{2}{j\omega}$ $\qquad \frac{a \cdot b}{c} = \text{ang}(a) + \text{ang}(b) - \text{ang}(c)$

$\frac{1}{2}\,\text{sgn}(t-1) \longleftrightarrow \frac{1}{2} e^{-j\omega} \cdot \frac{2}{j\omega}$

$\Rightarrow X(\omega) = \frac{1}{j\omega} e^{-j\omega}$

$= |X(\omega)| = \left|\frac{1}{j\omega}\right| \cdot |e^{-j\omega}| = \left|0 - \frac{j}{\omega}\right| \cdot 1 = \sqrt{0^2 + \frac{1}{\omega^2}} = \frac{1}{|\omega|}$

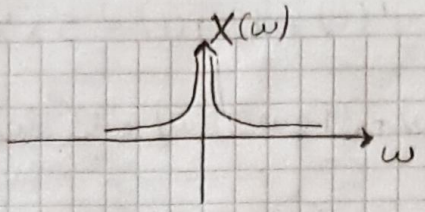
$\text{ang}\, X(\omega) = \text{ang}\left(\cos(-\omega) + j\sin(-\omega)\right) - \text{ang}\left(0 + j\omega\right)$

$= \tan^{-1}\left(\frac{\sin(-\omega)}{\cos(-\omega)}\right) - \tan^{-1}\left(\frac{\omega}{0}\right)$

$\text{ang}\, X(\omega) = \tan^{-1}(\tan(-\omega)) - \tan^{-1}\left(\frac{\omega}{0}\right) =$ $\begin{cases} -\omega - \frac{\pi}{2}, & \omega > 0 \\ -\omega - \left(-\frac{\pi}{2}\right), & \omega < 0 \end{cases}$

Bozulmasız iletim

$$\dot{X}(\omega) = \frac{1}{|\omega|}$$



10

Enerji Yoğunluğu Spektrumu

Örnek: $x(t) = \text{sinc}\left(\frac{t}{2}\right)$ a) $E_x = ?$ b) $\omega \in [-1/4, 1/4] \rightarrow E_x$

c) Toplam enerji = ?

$X(\omega) = 2\pi \cdot \text{rect}(\omega)$

a) $E_x = ?$

$$E_x = |X(\omega)|^2$$

$$E_x = |2\pi \cdot \text{rect}(\omega)|^2 = 4\pi^2 \cdot \text{rect}(\omega)$$

b) $\frac{1}{2\pi} \int_{-1/4}^{1/4} E_x \, d\omega \Rightarrow \frac{1}{2\pi} \int_{-1/4}^{1/4} 4\pi^2 \cdot \text{rect}(\omega) \, d\omega$

$$\frac{4\pi^2}{2\pi} \cdot \int_{-1/4}^{1/4} \text{rect}\,\omega \, d\omega = 2\pi \int_{-1/4}^{1/4} d\omega \qquad \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

c) $E = \frac{1}{2\pi} \int_{-\infty}^{\infty} 4\pi^2 \, \text{rect}\,\omega \, d\omega$ $\qquad 2\pi \cdot \frac{1}{2} = \pi$

$$\frac{4\pi^2}{2\pi} \int_{-1/2}^{1/2} d\omega \Rightarrow 2\pi \cdot \left(\frac{1}{2} + \frac{1}{2}\right) = 2\pi$$

Örnek: $7y''(t) + 11y'(t) + 13y(t) = 5x'(t) + 3x(t)$

$7(j\omega)^2 Y(\omega) + 11 j\omega Y(\omega) + 13Y(\omega) = 5 j\omega X(\omega) + 3X(\omega)$

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{3 + 5j\omega}{-7\omega^2 + 11j\omega + 13}$$

Örnek: $H(\omega) = \dfrac{-7\omega^2 + 11j\omega + 3}{-5\omega^2 + 2}$ sist. diferansiyel çiftisi nedir?

$$\frac{-7\omega^2 + 11j\omega + 3}{-5\omega^2 + 2} = \frac{Y(\omega)}{X(\omega)}$$

$$\Rightarrow -Y(\omega) 5\omega^2 + 2Y(\omega) = -7\omega^2 X(\omega) + 11 j\omega X(\omega) + 3X(\omega)$$

$$5(j\omega)^2 Y(\omega) + 2Y(\omega) = 7(j\omega)^2 X(\omega) + 11 j\omega X(\omega) + 3X(\omega)$$

$$5y''(t) + 2y(t) = 7x''(t) + 11x'(t) + 3x(t)$$

Örnek: $h(t) = 300\,\text{sinc}(300\pi t)$

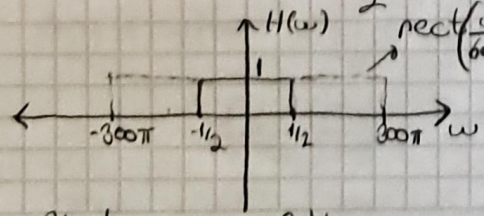
$x(t) = \frac{1}{2} + \frac{3}{4}\cos 200\pi t + \frac{1}{2}\cos 400\pi t - \frac{1}{4}\cos 600\pi t$

$y(t) = ?$ grafik çizmek frekans spektrumu

$$X(\omega) = \frac{2\pi}{2}\cdot\delta(\omega) + \frac{3\pi}{4}\left[\delta(\omega - 200\pi) + \delta(\omega + 200\pi)\right] + \frac{\pi}{2}\left[\delta(\omega - 400\pi) + \delta(\omega + 400\pi)\right] - \frac{\pi}{4}\left[\delta(\omega - 600\pi) + \delta(\omega + 600\pi)\right]$$

$\text{rect}(t) \leftrightarrow \text{sinc}\frac{\omega}{2}$ $\quad \text{sinc}\left(\frac{600\pi t}{2}\right) = \text{sinc}\,300\pi t$

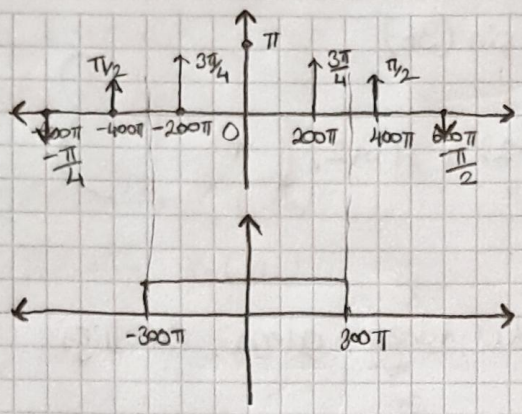
$2\pi\,\text{rect}(\omega) \longleftrightarrow \text{sinc}\frac{t}{2}$



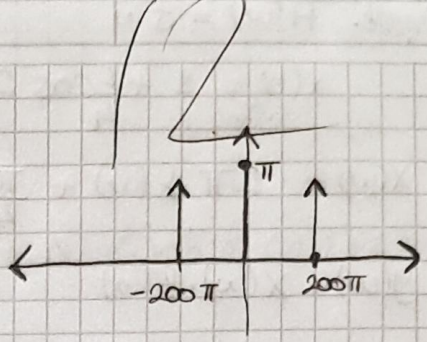
$\text{rect}\left(\frac{\omega}{600\pi}\right) \longleftrightarrow \frac{1}{600\pi}\,2\pi\,\text{rect}\left(\frac{\omega}{600\pi}\right)$

$H(\omega) = \text{rect}\left(\frac{\omega}{600\pi}\right)$

Alçak geçiren filtre



$\Rightarrow$



Örnek: $h(t) = \frac{2}{\pi}\,\text{sinc}(t)\cdot\cos(4t)$ $\qquad y(t) = ?$

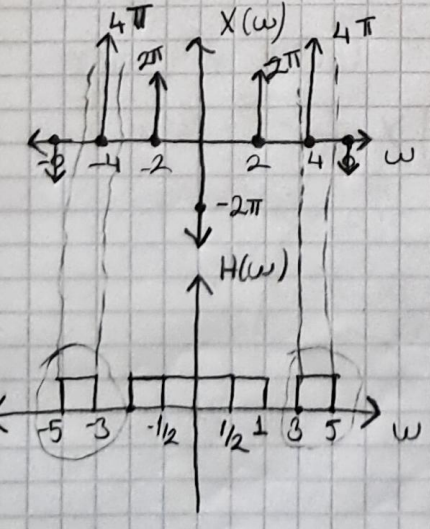
$x(t) = -1 + 2\cdot\cos 2t + 4\cdot\cos 4t - \cos 6t$

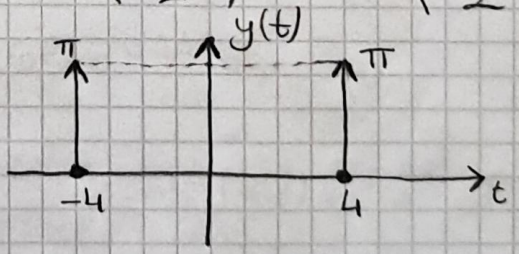
$X(\omega) = -2\pi\delta(\omega) + 2\pi\cdot[\delta(\omega-2)+\delta(\omega+2)] + 4\pi[\delta(\omega-4)+\delta(\omega+4)] - \pi[\delta(\omega-6)+\delta(\omega+6)]$

$h(t) = \frac{2}{\pi}\,\text{sinc}(t)\,\frac{1}{2}\left(e^{j4t} + e^{-j4t}\right)$

$h(t) = \underbrace{\frac{1}{\pi}\,\text{sinc}(t)\,e^{j4t}}_{V(\omega)} + \underbrace{\frac{1}{\pi}\cdot\text{sinc}(t)\cdot e^{-j4t}}_{V(\omega)}$

$H(\omega) = V(\omega-4) + V(\omega+4) = \text{rect}\left(\frac{\omega-4}{2}\right) + \text{rect}\left(\frac{\omega+4}{2}\right)$





$H(\omega) = \text{rect}\left(\frac{\omega-4}{2}\right) + \text{rect}\left(\frac{\omega+4}{2}\right)$

$\frac{1}{2}(\omega-4)$ $\qquad$ $\frac{1}{2}(\omega+4)$

Band geçiren filtre

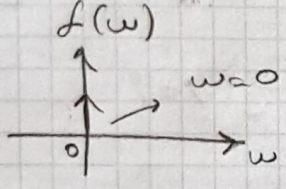
Örnek: $H(\omega) = j\omega$ $\quad y(t) = ?$

$$x(t) = 1 + \frac{1}{4}\cos 2t + \frac{1}{9}\sin(3t)$$

$$X(\omega) = 2\pi\,\delta(\omega) + \frac{\pi}{4}\left[\delta(\omega-2) + \delta(\omega+2)\right] + \left| \frac{}{} \right| 3$$

$$Y(\omega) = X(\omega)\cdot H(\omega)$$

$Y(\omega) = X(\omega)\cdot j\omega$ → zamanda türev alma özelliği



$$X(\omega) = 2\pi\,\delta(\omega) + \frac{\pi}{4}\left[\delta(\omega-2) + \delta(\omega+2)\right] + \frac{\pi}{9j}\left[\delta(\omega+3) - \delta(\omega-3)\right]$$

$$\cancel{j\omega\, 2\pi\,\delta(\omega)}^{0} + \frac{j\omega 2\pi}{4\cancel{2}}\delta(\omega-2) + j\omega\frac{\pi}{4}\delta(\omega+2) + \cancel{j\omega}\frac{3\pi}{9\cancel{j}}\delta(\omega+3) -$$

$$\cancel{j\omega}\frac{5\pi}{9\cancel{j}}\delta(\omega-3)\Big]$$

Fourier Dönüşümü

Doğrusallık Özelliği

$x_1(t) \xleftrightarrow{FT} X_1(\Omega)$

$x_2(t) \xleftrightarrow{FT} X_2(\Omega)$

$a\,x_1(t) + \beta\,x_2(t) \xleftrightarrow{FT} a\,X_1(\Omega) + \beta\,X_2(\Omega)$

$$\int_{-\infty}^{\infty} [a\,x_1(t) + \beta\,x_2(t)]\, e^{-j\Omega t}\, dt$$

! $$\int_{-\infty}^{\infty} e^{-j\Omega t}\, x(t)\, dt = a\underbrace{\int_{-\infty}^{\infty} x_1(t)\, e^{-j\Omega t} dt}_{X_1(\Omega)} + \beta\underbrace{\int_{-\infty}^{\infty} x_2(t)\, e^{-j\Omega t} dt}_{X_2(\Omega)}$$

Zamanda Ötelene

$x(t) \xleftrightarrow{FT} X(\Omega)$

$x(t-t_0) \xleftrightarrow{FT} e^{-j\Omega t_0}\, X(\Omega)$

$$\int_{-\infty}^{\infty} x(t-t_0)\, e^{-j\Omega t}\, dt$$

$t - t_0 = \ell$

$dt = d\ell$

$$\Rightarrow \int_{-\infty}^{\infty} x(\ell)\, e^{-j\Omega(\ell+t_0)}\, d\ell$$

$$= e^{-j\Omega t_0} \underbrace{\int_{-\infty}^{\infty} x(\ell)\, e^{-j\Omega \ell}\, d\ell}_{X(\Omega)}$$

Zamanda Ölçekleme

$x(t) \xleftrightarrow{FT} X(\Omega)$

$x(at) \xleftrightarrow{FT} \frac{1}{|a|} X\left(\frac{\Omega}{a}\right)$

$a > 0$

$$X(\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t} dt$$

$$\int_{-\infty}^{\infty} x(at)\, e^{-j\Omega t}\, dt$$

$at = \ell$

$a\,dt = d\ell$

$dt = \frac{d\ell}{a}$

$$\Rightarrow \frac{1}{a} \underbrace{\int_{-\infty}^{\infty} x(\ell)\, e^{\frac{-j\Omega \ell}{a}}\, d\ell}_{X\left(\frac{\Omega}{a}\right)}$$

$-\frac{1}{a} X\left(\frac{\Omega}{a}\right) \quad a < 0$

Eşlenik Özelliği

$x(t) \xleftrightarrow{FT} X(\Omega)$

$x^*(t) \xleftrightarrow{FT} \tilde{X}(\Omega)$

$\tilde{X}(\Omega) = X^*(\Omega)$

$$\tilde{X}(\Omega) = \int_{-\infty}^{\infty} x^*(t)\, e^{-j\Omega t}\, dt$$

$$\tilde{X}^*(\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{j\Omega t}\, dt$$

$$\tilde{X}^*(-\Omega) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j\Omega t}\, dt = X(\Omega)$$

Dualite Özelliği

$$X(\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t}\, dt \qquad x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\Omega)\, e^{j\Omega t}\, d\Omega$$

$t = \ell \qquad \Omega = \alpha$

$$x(\ell) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\alpha)\, e^{j\alpha\ell}\, d\alpha$$

$$2\pi \cdot x(\ell) = \int_{-\infty}^{\infty} X(\alpha)\, e^{j\alpha\ell}\, d\alpha$$

$\ell = -\ell$ için

$$2\pi\, x(-\ell) = \int_{-\infty}^{\infty} x(\alpha)\, e^{-j\alpha\ell}\, d\alpha$$

$$2\pi\, X(-\Omega) = \int_{-\infty}^{\infty} x(\alpha) \cdot e^{-j\Omega\alpha}\, d\alpha$$

$$2\pi\, X(-\Omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\Omega t}\, dt$$

Ayrık Zamanlı Sinyallerde Fourier Dönüşümü

$$x(n) \xleftrightarrow{DTFT} X(\Omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\Omega n}$$

Ters Fourier

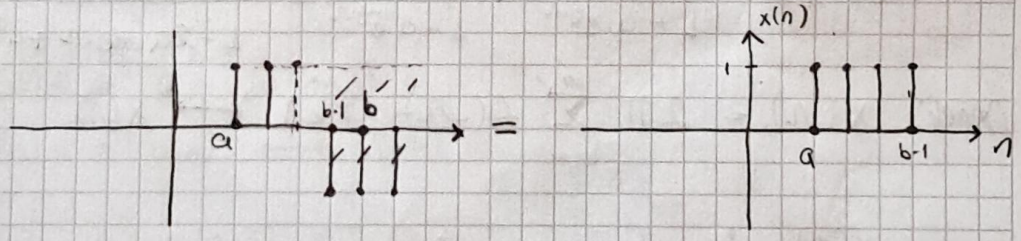
$$F^{-1} X(\Omega) = x(n) = \frac{1}{2\pi} \int_{2\pi} X(\Omega)\, e^{j\Omega n}\, d\Omega$$

Örnek: $x(n) = A \cdot \delta(n - n_0) \longleftrightarrow X(\Omega) = ?$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n) \cdot e^{-j\Omega n}$$

$$\Rightarrow \sum_{n=-\infty}^{\infty} A \cdot \delta(n-n_0) \cdot e^{-j\Omega n} = A \cdot e^{-j\Omega n} \Big|_{n=n_0} = A \cdot e^{-j\Omega n_0}$$

Örnek: $x(n) = u(n-a) - u(n-b)$, $b > a$ $\quad X(\Omega) = ?$



$$X(\Omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\Omega n} = \sum_{n=a}^{b-1} e^{-j\Omega n}$$

$n = n + a$ dönüşümü

$$= \sum_{n=0}^{b-a-1} e^{-j\Omega(n+a)} = e^{-j\Omega a} \sum_{n=0}^{b-a-1} (e^{-j\Omega})^n = e^{-j\Omega a} \cdot \frac{1 - e^{-j\Omega(b-a)}}{1 - e^{-j\Omega}}$$

$$= \frac{e^{-j\Omega a} - e^{-j\Omega b}}{1 - e^{-j\Omega}}$$

$$= \frac{e^{-j\Omega\left(\frac{a+b}{2}\right)} \left[e^{-j\Omega\left(\frac{a-b}{2}\right)} - e^{j\Omega\left(\frac{a-b}{2}\right)} \right] / 2j}{e^{-j\frac{\Omega}{2}} \left[e^{j\frac{\Omega}{2}} - e^{-j\frac{\Omega}{2}} \right] / 2j}$$

$$= -e^{-j\Omega\left(\frac{a+b-1}{2}\right)} \frac{\sin\left(\frac{a-b}{2}\Omega\right)}{\sin\left(\frac{\Omega}{2}\right)} = e^{-j\Omega\left(\frac{a+b-1}{2}\right)} \frac{\sin\left(\frac{b-a}{2}\right)\Omega}{\sin\left(\frac{\Omega}{2}\right)}$$

Örnek: $x(n) = a^n u(n) \longleftrightarrow X(\Omega) \quad |a| < 1$

$$X(\Omega) = \sum_{-\infty}^{\infty} x(n)\, e^{-j\Omega n} = \sum_{n=0}^{\infty} a^n e^{-j\Omega n}$$

$$= \sum_{0}^{\infty} \left(a \cdot e^{-j\Omega}\right)^n = \frac{1}{1 - a \cdot e^{-j\Omega}} = \frac{e^{j\Omega}}{e^{j\Omega} - a}$$

sonsuz toplam formülü

Örnek: $x(n) = a^{|n|}, \quad |a| < 1, \quad X(\Omega) = ?$

$$X(\Omega) = \sum_{n=-\infty}^{\infty} a^{|n|} \cdot e^{-j\Omega n} = \sum_{n=-\infty}^{-1} a^{-n} \cdot e^{-j\Omega n} + \sum_{n=0}^{\infty} a^n e^{-j\Omega n}$$

$$= \sum_{n=1}^{\infty} a^n \cdot e^{j\Omega n} + \sum_{n=0}^{\infty} \left(a \cdot e^{j\Omega}\right)^n$$

$$= X(\Omega) = \frac{a \cdot e^{j\Omega}}{1 - a \cdot e^{j\Omega}} + \frac{1}{1 - a \cdot e^{-j\Omega}} = \frac{1 - a^2}{1 - 2a\cos\Omega + a^2}$$

Örnek: $X(\Omega) = 2\pi \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k) \longleftrightarrow x(n)$

$$x(n) = \frac{1}{2\pi} \int_{\langle 2\pi \rangle} X(\Omega)\, e^{j\Omega n}\, d\Omega$$

$$x(n) = \frac{1}{\cancel{2\pi}} \int_{-\pi}^{\pi} \cancel{2\pi} \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)\, e^{j\Omega n}\, d\Omega$$

$$= \int_{-\pi}^{\pi} \delta(\Omega)\, e^{j\Omega n}\, d\Omega = e^{j\Omega n}\Big|_{\Omega=0} = 1$$

Erdem

Frekansta tersi alma $n \cdot x(n) \longleftrightarrow j \frac{d\,X(\Omega)}{d\Omega}$ $\quad \frac{1}{M} X\left(\frac{\Omega - 2\pi k}{M}\right)$

$x(n/M) \longleftrightarrow X(M \cdot \Omega)$

Örnek: $\delta(n) \overset{DTFT}{\longleftrightarrow} 1$

$$u(n) \longleftrightarrow \frac{e^{j\Omega}}{e^{j\Omega}-1} + \pi \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)$$

$$x(n) = 2\delta(n) - u(n) \overset{DTFT}{\longleftrightarrow} X(\Omega)$$

$$2.\ F\{\delta(n)\} - F\{u(n)\}$$

$$2 - \left(\frac{e^{j\Omega}}{e^{j\Omega}-1} + \pi \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k)\right)$$

$$= \frac{e^{j\Omega} - 2}{e^{j\Omega} - 1} + \text{---}$$

Örnek: $a^n u(n) \overset{DTFT}{\longleftrightarrow} \frac{1}{1 - a \cdot e^{-j\Omega}}$

$$x(n) = a^n \cdot u(n-3) \overset{DTFT}{\longleftrightarrow} X(\Omega)$$

$$x(n) = a^3 \cdot a^{n-3} \cdot u(n-3)$$

$$v(n) = a^n u(n) \longleftrightarrow V(\Omega) = \frac{1}{1 - a \cdot e^{-j\Omega}}$$

$$x(n) = a^3 \cdot v(n-3)$$

$$v(n-3) \longleftrightarrow e^{-j3\Omega} \cdot \frac{1}{1 - a e^{-j\Omega}} \qquad X(\Omega) = a^3 \cdot \frac{e^{-j3\Omega}}{1 - a \cdot e^{-j\Omega}}$$

Örnek: $x(n) = \cos(\omega_0 n) \xleftrightarrow{\text{DTFT}} X(\omega) = ?$

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$$x(n) = \frac{1}{2}\left(1 \cdot e^{j\omega_0 n} + 1 \cdot e^{-j\omega_0 n}\right)$$

$$1 \xleftrightarrow{\text{DTFT}} 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k) = Y(\omega)$$

$$X(\omega) = \frac{1}{2}\left[Y(\omega - \omega_0) + Y(\omega + \omega_0) \right]$$

$$= \pi \cdot \sum_{k=-\infty}^{\infty} \left[\delta(\omega - \omega_0 - 2\pi k)\ \delta(\omega + \omega_0 - 2\pi k) \right]$$

Örnek: $x(n)$ reel ise $X(\omega)$'nın eşlenik simetrik olduğunu gösteriniz

$$\mathcal{F}\{x^*(n)\} = X^*(n)$$

x reel ise $X(n) = x^*(n)$

$$\mathcal{F}\{x(n)\} = X^*(-\omega) \Rightarrow X(\omega) = X^*(-\omega)$$

Örnek: $x(n) = a^{|n|} \cdot u(n) \longleftrightarrow X(\omega)$

$$x(n) = a^n \cdot u(n) + a^{-n} \cdot u(-n) - \delta(n)$$

$$= \frac{1}{1 - a \cdot e^{-j\omega}} + \frac{1}{1 - a \cdot e^{j\omega}} - 1$$

$$= \frac{1 - a^2}{1 - 2a\cos\omega + a^2}$$

Örnek: $x(n) = u(n) \xleftrightarrow{DTFT} \frac{1}{1-e^{-j\Omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(\Omega - 2\pi k) = X(\Omega)$

$y(n) = (\uparrow 3)\, x(n) \xleftrightarrow{DTFT} Y(\Omega) = ?$

$(\uparrow 3)\, x(n) = x\left(\frac{1}{3}n\right) \longleftrightarrow X(3\Omega) = Y(\Omega)$

$Y(\Omega) = \frac{1}{1-e^{-j3\Omega}} + \pi \sum_{k=-\infty}^{\infty} \delta(3\Omega - 2\pi k)$

Örnek: $x(n) = \frac{1}{4}\,\mathrm{sinc}\left(\frac{\pi}{4}n\right)$

$y(n) = (\downarrow 2)\, x(n) = x(2n) \xleftrightarrow{DTFT} Y(\Omega) = ?$

$\frac{B}{\pi}\,\mathrm{sinc}(Bn) \longleftrightarrow \sum_{k=-\infty}^{\infty} \mathrm{rect}\left(\frac{\Omega - 2\pi k}{2B}\right)$

$B = \frac{\pi}{4} \quad \frac{B}{\pi} = \frac{1}{4}$

$X(\Omega) = \sum_{k=-\infty}^{\infty} \mathrm{rect}\left(\frac{2}{\pi}(\Omega - 2\pi k)\right)$

$M = 2$

$Y(\Omega) = \frac{1}{M} \sum_{k=0}^{M-1} X\left(\frac{\Omega - 2\pi k}{M}\right) = \frac{1}{2} \sum_{k=0}^{1} X\left(\frac{\Omega - 2\pi k}{2}\right)$

$= \frac{1}{2} X\left(\frac{\Omega}{2}\right) + \frac{1}{2} X\left(\frac{\Omega - 2\pi}{2}\right)$

$= \frac{1}{2} \sum_{k=-\infty}^{\infty} \mathrm{rect}\left(\frac{2i}{\pi} \cdot \frac{\Omega}{2} - 2\pi k\right)$

Konvolüsyon özelliği örnek: $x_1(n) = a^n u(n)$

$x_2(n) = a^{|n|}$, $|a| < 1$

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$x(n) = x_1(n) * x_2(n) = X(\Omega) = ?$

$X(\Omega) = X_1(\Omega) \cdot X_2(\Omega)$

$X_1(\Omega) = \frac{1}{1 - a.e^{-j\Omega}}$

$X_2(\Omega) = \frac{1 - a^2}{1 - 2a\cos\Omega + a^2}$

$\Rightarrow \left(\frac{1}{1 - a.e^{-j\Omega}}\right) \cdot \left(\frac{1 - a^2}{1 - 2a.\cos\Omega + a^2}\right)$

Örnek: Çarpma özelliği

$x(n) = \frac{B}{2\pi} \text{sinc}^2\left(\frac{B}{2} n\right) \xleftrightarrow{DTFT} X(\Omega) = ?$

$\frac{B}{\pi} \text{sinc}(Bn) \xleftrightarrow{DTFT} \sum_{k=<\infty>} \text{rect}\left[\frac{\Omega - 2\pi k}{2B}\right]$

$B = \frac{B}{2}$ için

$\frac{B}{2\pi} \text{sinc}\left(\frac{B}{2} n\right) \xleftrightarrow{DTFT} \sum_{k=<\infty>} \text{rect}\left(\frac{\Omega - 2\pi k}{B}\right)$

$\underbrace{\frac{B}{2\pi} \text{sinc}\left(\frac{B}{2} n\right)}_{V(n)} \cdot \underbrace{\text{sinc}\left(\frac{B}{2} n\right)}_{V(n)} \longleftrightarrow \frac{B}{2\pi} \int_{2\pi} V(m) \cdot V(\Omega - m)\, dm$

$v(n) = \text{sinc}\left(\frac{B}{2} n\right) \longleftrightarrow V(\Omega) = \frac{2\pi}{B} \sum_{k=<\infty>} \text{rect}\left(\frac{\Omega - 2\pi k}{B}\right)$

$= \frac{B}{4\pi^2} \int_{-\pi}^{\pi} \text{rect}\left(\frac{\Omega}{B}\right) \cdot \text{rect}\left(\frac{\Omega - m}{B}\right) d\Omega$ olur

$X(\Omega) = \text{tri}\left(\frac{\Omega}{2B}\right)$ tek periyot için

Önnek: $x(n) = n \cdot a^n \, u(n) \xleftrightarrow{DTFT} X(\Omega)$ Frekansta türev

$x(n) = n \cdot v(n)$

$v(n) = a^n \cdot u(n) \xleftrightarrow{DTFT} V(\Omega) = \frac{1}{1 + a \cdot e^{-j\Omega}}$

$X(\Omega) = j \frac{dV(\Omega)}{d\Omega} = \frac{a \cdot e^{-j\Omega}}{(1 - a \cdot e^{-j\Omega})^2}$

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Önnek: $x(n) = a^{|n|} - a^{|n-1|} \xleftrightarrow{DTFT} X(\Omega) = ?$

$a^{|n|} \xleftrightarrow{DTFT} \frac{1 - a^2}{1 - 2 \cdot a \cdot \cos\Omega + a^2}$

$X(\Omega) = (1 - e^{-j\Omega}) \frac{1 - a^2}{1 - 2a\cos\Omega + a^2}$

Önnek: $x(n) = u(n) \xleftrightarrow{DTFT} X(\Omega) = ?$

$u(n) = \sum_{k=-\infty}^{n} \delta(k)$

$\delta(n) \longleftrightarrow 1 = \delta(\Omega)$

Σ

Parseval

Örnek: $x(n) = \text{sinc}\left(\frac{1}{2}n\right)$

$B = \frac{1}{2} \rightarrow 2\pi \cdot \frac{1}{2\pi} \text{sinc}(Bn) = x(n)$

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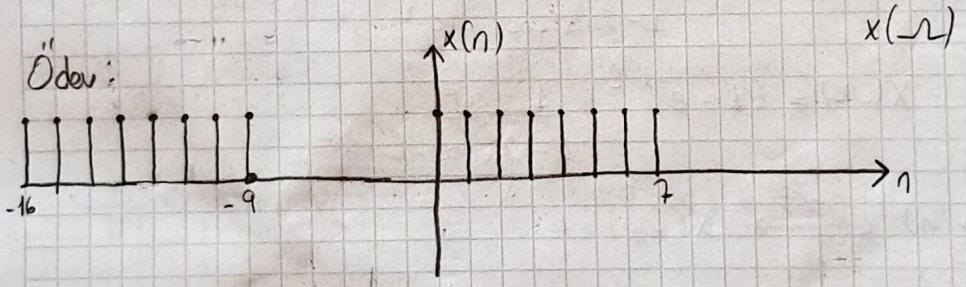
$X(\Omega) = 2\pi \cdot \sum_{k=<\infty>} \text{rect}(\Omega - 2\pi k)$

$E = \frac{1}{2\pi} \int_{2\pi} |X(\Omega)|^2 d\Omega = \frac{1}{2\pi} \int_{-\pi}^{\pi} (2\pi)^2 \text{rect}^2(\Omega) =$

$= 2\pi \int_{-1/2}^{1/2} \text{rect}\,\Omega \, d\Omega = 2\pi$

Ödev:



11

FT Derste işlenenler

Örnek: $x(n) = \left(\frac{1}{2}\right)^{|n|}$

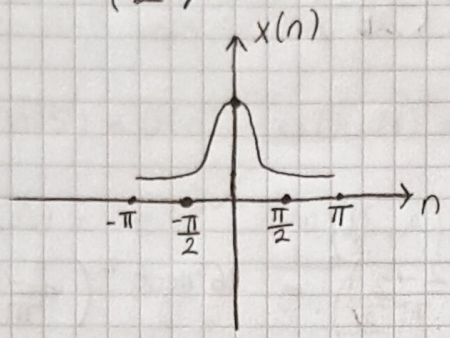
$$X(\Omega) = \frac{1 - a^2}{1 - 2a\cos\Omega + a^2}, \quad a = 1/2$$

$$= \frac{3}{5 - 4\cos\Omega}$$

$\frac{3}{5} = 0.6$

$\frac{3}{9} = \frac{1}{3} = 0.33$

$\Omega \in (-\pi, \pi]$



Örnek: $x(n) = \left(\frac{1}{2}\right)^n u(n)$

$$X(\Omega) = \frac{1}{1 - \frac{1}{2}e^{-j\Omega}}$$

$$|X(\Omega)| = \left| \frac{1}{1 - \frac{1}{2}\left(\cos(-\Omega) + j\sin(-\Omega)\right)} \right|$$

$$= \frac{1}{\left|1 - \frac{1}{2}\cos(\Omega) + \frac{1}{2}j\sin(-\Omega)\right|}$$

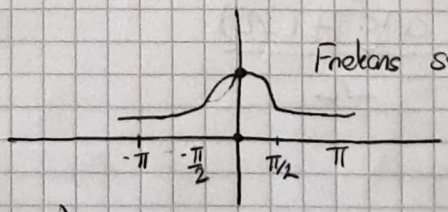
$$|X(\Omega)| = \frac{1}{\sqrt{1 + \frac{1}{4}\cos^2\Omega - \cos\Omega + \frac{1}{4}\sin^2\Omega}}$$

$$= \frac{1}{\sqrt{\left(1 - \frac{1}{2}\cos(\Omega)\right)^2 + \left(\frac{1}{2}\sin(\Omega)\right)^2}}$$

$$X(\Omega) = \frac{1}{\sqrt{\frac{5}{4} - \cos\Omega}}$$

$$= \frac{2}{\sqrt{5 - 4\cos\Omega}}$$

Frekans spektrumu



$$\arg\left(X(\Omega)\right) = \arg\left(\frac{1}{1 - \frac{1}{2}e^{-j\Omega}}\right) = \arg(1) - \arg\left(1 - \frac{1}{2}e^{-j\Omega}\right)$$

$$= 0 - \arg\left(1 - \frac{1}{2}\cos\Omega + \frac{1}{2}j\sin\Omega\right)$$

$$-\tan^{-1}\left(\frac{\frac{1}{2}\sin\Omega}{1 - \frac{1}{2}\cos\Omega}\right) = -\tan^{-1}\left(\frac{\sin\Omega}{2 - \cos\Omega}\right)$$

Faz spektrumu

Enerji yoğunluğu spektrumu örnek soru

a) $E_x = ?$

b) $\Omega \in [\frac{-\pi}{8} , \frac{\pi}{8}]$

$$x(n) = \text{sinc}\ \frac{\pi}{4} n$$

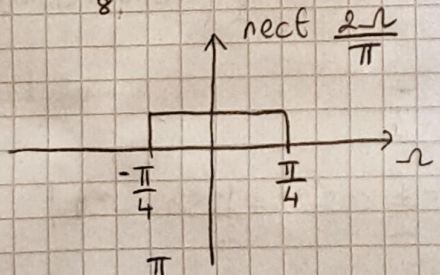
$$\text{sinc}\ Bn \longleftrightarrow \frac{\pi}{B}\ \text{rect}\left(\frac{\Omega}{2B}\right)$$

a) $B = \frac{\pi}{4}$

$$X(\Omega) = 4 . \text{rect}\left(\frac{2\Omega}{\pi}\right)$$

$$E_x = |X(\Omega)|^2 = \left(4\ \text{rect}\ \frac{2\Omega}{\pi}\right)^2 = 16\ \text{rect}^2\left(\frac{2\Omega}{\pi}\right)$$

b) $$E_1 = \frac{1}{2\pi} \int_{\frac{-\pi}{8}}^{\pi/8} E_x\, d\Omega = \frac{1}{2\pi} \int_{\frac{-\pi}{8}}^{\pi/8} 16\ \text{rect}^2\left(\frac{2\Omega}{\pi}\right) d\Omega$$



$$E_1 = \frac{8}{\pi} \int_{\frac{-\pi}{8}}^{\frac{\pi}{8}} 1\, d\Omega = 2$$

$$E_{toplam} = \frac{1}{2\pi} \int_{-\pi}^{\pi} 16\ \text{rect}^2\left(\frac{2\Omega}{\pi}\right) d\Omega = \frac{8}{\pi} \int_{\frac{-\pi}{4}}^{\pi/4} d\Omega = 4\ \text{joule}$$

$$\delta\ \tau_p = -\frac{\arg H(\Omega)}{\Omega}$$

Örnek: $h(n) = u(n+5) - u(n-6)$

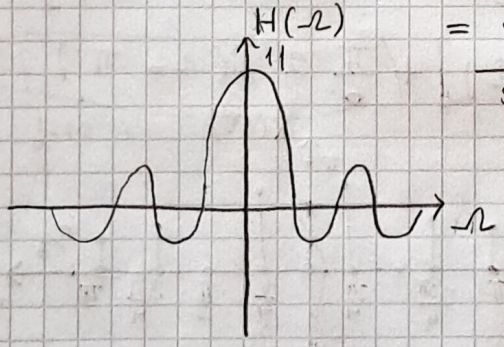
$n = n-5$ için

$h(n-5) = u(n) - u(n-11) = v(n)$

$V(\Omega) = e^{j5\Omega} \dfrac{\sin\left(\frac{11}{2}\Omega\right)}{\sin\left(\frac{\Omega}{2}\right)}$

$v(n) = h(n-5)$

$h(n) = v(n+5) \longleftrightarrow H(\Omega) = e^{j5\Omega} \cdot e^{-j5\Omega} \left(\dfrac{\sin \frac{11}{2}\Omega}{\sin \frac{\Omega}{2}} \right)$



$= \dfrac{\sin \frac{11}{2}\Omega}{\sin \frac{\Omega}{2}}$

Fark Denklemleri Örnek

$5y(n) + 2y(n-1) + 3y(n-2) = x(n) - 2x(n-1)$, $H(\Omega) = ?$

$5y(\Omega) + 2e^{-j\Omega} y(\Omega) + 3e^{-j2\Omega} y(\Omega) = x(\Omega) - 2.e^{-j\Omega} X(\Omega)$

$$H(\Omega) = \frac{y(\Omega)}{X(\Omega)} = \frac{1 - 2e^{-j\Omega}}{5 + 2.e^{-j\Omega} + 3.e^{-j2\Omega}}$$

Örnek: $h(n) = \frac{1}{3} \sin\left(\frac{\pi}{3} n\right)$

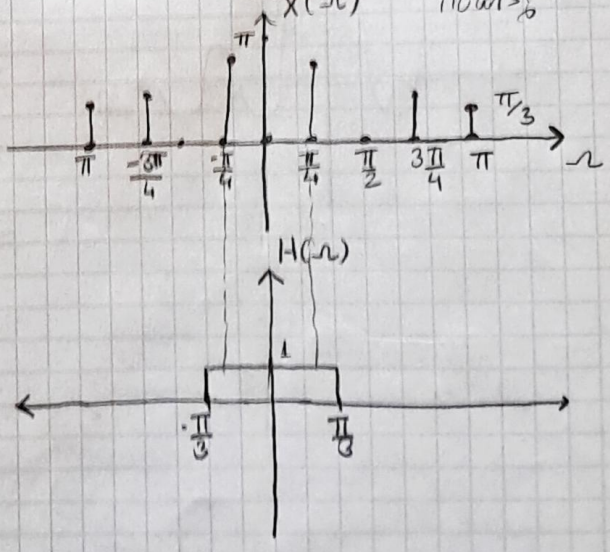
$$x(n) = \frac{1}{2} + \frac{2}{3}\cos\left(\frac{\pi}{4} n\right) + \frac{1}{2}\cos\left(\frac{3\pi}{4} n\right) + \frac{1}{6}\cos(\pi n)$$

$\Omega \in (-\pi, \pi)$ $y(n) = ?$

$$X(\Omega) = \frac{1}{2} 2\pi \delta(\Omega) + \frac{2\pi}{3}\left[\delta\left(\Omega - \frac{\pi}{4}\right) + \delta\left(\Omega + \frac{\pi}{4}\right)\right] +$$

$$\frac{\pi}{2}\left[\delta\left(\Omega - \frac{3\pi}{4}\right) + \delta\left(\Omega + \frac{3\pi}{4}\right)\right] + \frac{\pi}{6} 2.\delta(\Omega - \pi) + \delta(\Omega + \pi)$$

$$H(\Omega) = \text{rect}\left(\frac{3\Omega}{2\pi}\right)$$



Onde: $h(n) = \frac{8}{5\pi} \sin\left(\frac{4}{5}n\right) \cdot \cos\left(\frac{6}{5}n\right)$

$$x(n) = \frac{1}{4} + \cos\left(\frac{4}{5}n\right) + \frac{2}{3}\cos\frac{8}{5}n + \frac{1}{4}\cos\frac{12}{5}n + \frac{1}{6}\cos\pi n$$

Laplace Dönüşümü

$$\mathcal{L}\{x(t)\} = \int_{-\infty}^{\infty} x(t)\cdot e^{-st}\,dt$$

$$\mathcal{L}^{-1}\{X(s)\} = \frac{1}{j2\pi}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)\,e^{st}\,ds$$

$s = \sigma + j\omega$

* Sistemlerin kararlılığını ölçmek için laplace dönüşümü kullanılır.

Önce: $x(t) = e^{-at}u(t)$ $\quad a \in R$

$X(s) = \mathcal{L}\{x(t)\} = ?$

$$\int_{-\infty}^{\infty} e^{-at}e^{-st}u(t)\,dt$$

$$X(s) = \int_{-\infty}^{\infty} x(t)\cdot e^{-st}\,dt \qquad = \int_{0}^{\infty} e^{-(s+a)t}\,dt$$

$$= \frac{-1}{s+a}\cdot e^{-(s+a)t}\Big|_0^{\infty} \Rightarrow \frac{-1}{s+a}\left[e^{-(s+a)\infty} - \underbrace{e^{-(s+a)0}}_{1}\right]$$

$s+a>0$ ise $e^{-(s+a)\cdot\infty} = 0$

$s+a<0$ ise $e^{-(s+a)\cdot\infty} = \infty$

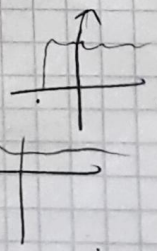
$$X(s) = \frac{-1}{s+a}[0-1] = \frac{1}{s+a}$$

$Re(s) > -a$

↓

yakınsama bölgesi

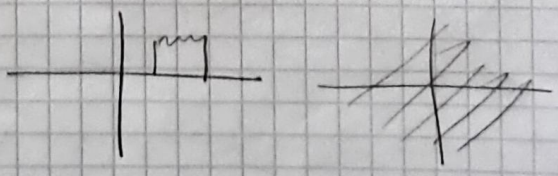
ROC



Fonksiyon sağ taraflıysa ROC da sol taraflıdır.

Fonksiyon çift taraflıysa ROC bir şerittir

Fonksiyon belli bir aralıktaysa ROC her yerdir



Örnek: $x(t) = -e^{-at}\,u(-t)$ ise $X(s)$ nedir?

ROC nedir?

$$X(s) = \int_{-\infty}^{\infty} x(t)\cdot e^{-st}\,dt$$

$$\Rightarrow X(s) = -\int_{-\infty}^{0} e^{-at}\cdot e^{-st}\,dt = -\int_{-\infty}^{0} e^{-(s+a)t}\,dt = \frac{1}{s+a}\left(e^{-(s+a)t}\Big|_{-\infty}^{0}\right)$$

$$= \frac{1}{s+a}\left(e^{-(s+a)\cdot 0} - e^{-(s+a)\cdot(-\infty)}\right)$$

$e^{-(s+a)t}$

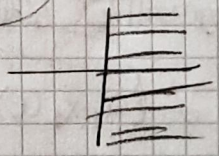
$e^{-at}\,u(t) \leftrightarrow \frac{1}{s+a}$ $\quad Re(s) > -a$

$-e^{-at}\,u(-t) \leftrightarrow \frac{1}{s+a}$ $\quad Re(s) < -a$

Örnek: $x(t) = u(t)$, $X(s)$, ROC = ?

$$X(s) = \int_{-\infty}^{\infty} x(t)\cdot e^{-st}\,dt$$

$$\Rightarrow X(s) = \int_{0}^{\infty} e^{-st}\,dt = \frac{-1}{s}\,e^{-st}\Big|_{0}^{\infty} = \frac{1}{s} \qquad Re(s) > 0$$



Örnek: $X(s) = \dfrac{s+\frac{1}{2}}{(s^2+2s+2)(s^2+s-2)}$

ROC=?

$s^2+2s+2=0$

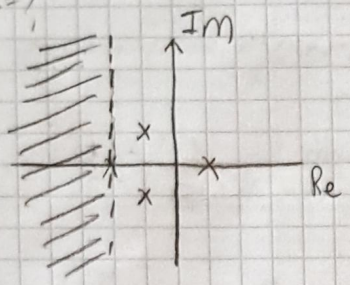
$s_1 = -1+j$

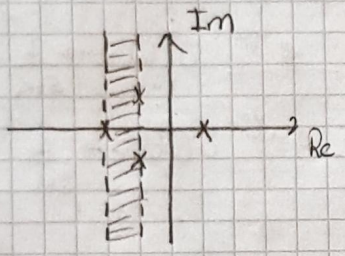
$s_2 = -1-j$

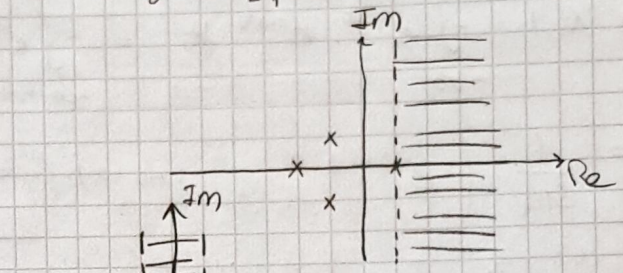
$s^2+s-2=0$ $\quad s_1=-2$ $\quad s_2=1$

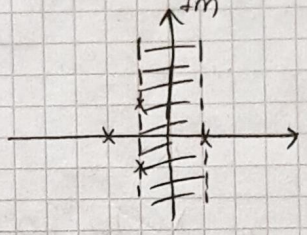
$s \quad +2$

$s \quad -1$









$X(s) = \dfrac{N(s)}{D(s)}$ N ve D aynı tipte fonksiyonsa X nasyonel'dir aynı tipten basit polinom, trigonometrik veya üstel olabilir.

Doğrusallık özelliği

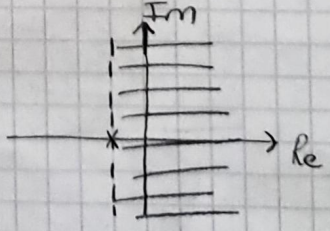
$$\mathcal{L}\{a_1 \cdot x_1(t) + a_2 \cdot x_2(t)\} = a_1 \cdot X_1(s) + a_2 \cdot X_2(s) \quad ROC\ R_1 \cap R_2$$

Örnek: $x_1(t) = e^{-t} \cdot u(t)$, $x_2(t) = e^{-t} u(t) - e^{-2t} \cdot u(t)$

$x(t) = x_1(t) + x_2(t)$

$X_1(s) = \dfrac{1}{s+1}$, $Re(s) > -1$

$X_2(s) = \dfrac{1}{s+1} - \dfrac{1}{s+2}$



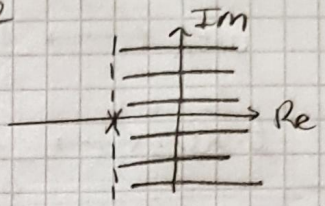
$Re(s) > -1 \cap Re(s) > -2 = Re(s) > -1$

$X(s) = \dfrac{1}{s+1} + \dfrac{1}{s+1} - \dfrac{1}{s+2} = \dfrac{2}{s+1} - \dfrac{1}{s+2} = \dfrac{s+3}{(s+1)\cdot(s+2)}$ $\quad Re(s) > -1$

$x(t) = x_1(t) - x_2(t)$

$X(s) = X_1(s) - X_2(s) = \frac{1}{s+1} - \frac{1}{s+1} + \frac{1}{s+2} = \frac{1}{s+2}$

$Re(s) > -2$

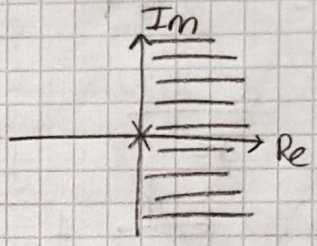


Zamanda kaydırma

Örnek: $x(t) = u(t-1)$

$u(t) \overset{LT}{\longleftrightarrow} \frac{1}{s}$, $Re(s) > 0$

$u(t-1) \longleftrightarrow e^{-s} \cdot \frac{1}{s}$, $Re(s) > 0$

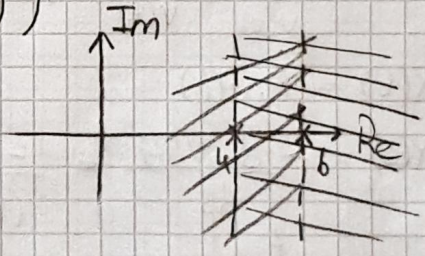


Örnek: $x(t) = e^{5t} \cdot e^{-|t|} = e^{5t} \cdot e^{-t} \cdot u(t) + e^{5t} \cdot e^{t} \cdot u(-t)$

$x(t) = e^{4t} \cdot u(t) - \left(-e^{6t} \cdot u(-t)\right)$

$X(s) = \frac{1}{s-4} - \frac{1}{s-6}$

$Re(s) > 4 \quad Re(s) < 6$

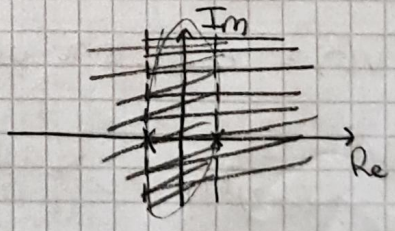


Örnek: $x(t) = e^{-|t|}$, $y(t) = e^{-3|t|}$

$x(t) = e^{-t} u(t) - \left(-e^{t} u(-t)\right)$

$X(s) = \frac{1}{s+1} - \frac{1}{s-1}$

$Re(s) > -1 \quad Re(s) < 1$



$y(t) = x(3t)$

$y(s) = \frac{1}{|3|} \cdot X(s/3) = \frac{1}{3} \cdot \frac{-2}{\left(\frac{s}{3}\right)^2 - 1}$ $\quad Re\left(\frac{s}{3} > -1\right) \cap Re\left(\frac{s}{3} < 1\right)$

konvolüsyon

Örnek: $x_1(t) = \sin 3t \cdot u(t)$, $x_2(t) = t \cdot u(t)$

$\sin 3t\, u(t) \longleftrightarrow \frac{3}{s^2+9}$ $\quad Re(s) > 0$ (sin için)

$u(t) \longleftrightarrow \frac{1}{s}$ $\quad Re(s) > 0$

$t \cdot u(t) \longleftrightarrow \frac{1}{s^2}$, $Re(s) > 0$

$x_1(t) * x_2(t) = y(t)$

$Y(s) = X_1(s) \cdot X_2(s)$

$= \frac{1}{s^2} \cdot \frac{3}{s^2+9} = \frac{3}{s^2(s^2+9)}$ $\quad Re(s) > 0$

Örnek: $\frac{d\, f(t)}{dt}$

$\frac{d\, f(t)}{dt} \longleftrightarrow s\, F(s) = s \cdot 1 = s$, $\forall s$

$f(t) \xleftrightarrow{LT} 1 \quad F(s)$

Örnek: $x(t) = t \cdot \underbrace{e^{-2t} u(t)}_{v(t)}$ $\qquad V(s) = \frac{1}{s+2}$ $\quad Re(s) > -2$

$x(t) = t \cdot v(t)$

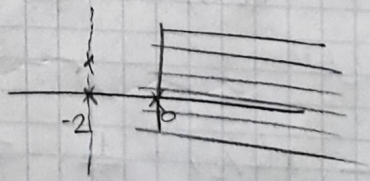
$X(s) = -\frac{d\, V(s)}{ds} = \frac{-(-1)}{(s+2)^2} = \frac{1}{(s+2)^2}$, $Re(s) > -2$

Örnek: $x(t) = \int_{-\infty}^{t} e^{-2\tau} \cdot \sin\tau\, u(\tau)\, d\tau$ $\qquad$ Zamanda integral alma

$v(t) = e^{-2t} \cdot \sin t\, u(t)$ $\qquad (s+2)^2 = -1$

$V(s) = \frac{1}{(s+2)^2+1}$, R_1 ; $Re(s) > -2$ $\qquad s+2 = j$

$s = -2 + j$



$x(t) = \int_{-\infty}^{t} v(\tau)\, d\tau$

$X(s) = \frac{1}{s} V(s) \Rightarrow \frac{1}{s[(s+2)^2+1]}$ $\quad R_1 \cap Re(s) > 0$

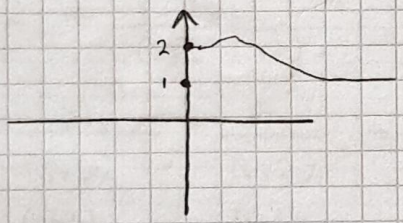
Örnek: Sınırlı değere sahip ve nedensel fonksiyonun laplace dönüşümü şu şekilde veriliyor:

$$X(s) = \frac{2s^2 + 3s + 2}{s^3 + 2s^2 + 2s}, \quad Re(s) > 0$$

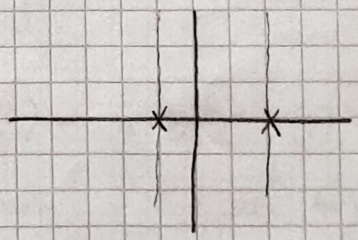
$x(0) = ?$ $\quad x(\infty) = ?$

$$x(0) = \lim_{s\to\infty} s.X(s) = \lim_{s\to\infty} \frac{2s^2+3s+2}{s^2+2s+2} = 2$$

$$x(\infty) = \lim_{s\to 0} s.X(s) = \lim_{s\to 0} \frac{2s^2+3s+2}{s^2+2s+2} = 1$$



Örnek: $X(s) = \frac{2}{s^2-s-2}$, ROC : $-1 < Re(s) < 2$



$$X(s) = \frac{2}{(s+1).(s+2)} = \frac{A_1}{(s+1)} + \frac{A_2}{(s-2)}$$

$$\frac{A_1}{(s-2)} \quad \frac{A_2}{(s+1)}$$

$$A_1 s - 2A_1 + A_2 s + A_2 = 2$$

$$A_1 + A_2 = 0$$
$$-/ \; -2A_1 + A_2 = 2$$
$$3A_1 = -2$$
$$A_1 = \frac{-2}{3}$$
$$A_2 = \frac{2}{3}$$

$$\frac{-2}{3}\cdot\frac{1}{s+1} + \frac{2}{3}\cdot\frac{1}{s-2}$$

$e^{-t}.u(t)$ $\quad -e^{-t}.u(-t)$

$e^{2t}.u(t)$ $\quad -e^{+2t}.u(-t)$

$$x(t) = \frac{-2}{3}e^{-t}.u(t) + \frac{2}{3}e^{2t}.u(-t)$$

Önrdi. $X(s) = \frac{2s+1}{(s+1)^2 \cdot (s+2)}$, $Re(s) > -1$ $x(t) = ?$

$$x(t) = 3 \cdot e^{-t} \cdot u(t) - 6 \cdot e^{-t} \cdot u(t) - 3 \cdot e^{-2t} \cdot u(t)$$

$$X(s) = \frac{k_1}{s+1} + \frac{k_2}{(s+1)^2} + \frac{A_1}{s+2}$$

$$k_1 = \frac{1}{(2-1)!_0} \left(\frac{d}{ds}\right)^{(2-1)} \cancel{(s+1)^2} \cdot \frac{(2s+1)}{\cancel{(s+1)^2} \cdot (s+2)} \Bigg|_{s=-1}$$

$$k_1 = \left[\frac{d}{ds}\left(\frac{2s+1}{s+2}\right)\right]_{s=-1} = \frac{2(s+2) - (2s+1)}{(s+2)^2} \Bigg|_{s=-1} = 3$$

$$k_2 = \frac{1}{(2-2)!_0} \left(\frac{d}{ds}\right)^{2-2} (s+1)^2 \cdot$$

Örnek: $X(s) = \frac{s+5}{s^2+2s+2}$ $\quad Re(s) > 0$

$\mathcal{L}^{-1}\{X(s)\} \Rightarrow X(s) = \frac{s+1}{(s+1)^2+1} + \frac{4}{(s+1)^2+1}$

$x(t) = e^{-t}\cdot\cos t\cdot u(t) + 4\cdot e^{-t}\sin t\, u(t)$

Ek Derste işlenenler 25.12.2023

Örnek:

$X(s) = \frac{2s^2+4s+5}{(s+1)\cdot(s+2)}$, $Re(s) > -1$

! Bir sistem nedensel ise ROC da sağ taraflıdır.

$X(s) = 2 + \frac{(-2s+1)}{(s+1)\cdot(s+2)}$

$= \frac{2}{1} + \frac{A_1}{(s+1)} + \frac{A_2}{(s+2)} = 2\cdot(s+1)(s+2)$

$A_1 = 3$, $A_2 = -5$

$\mathcal{L}^{-1}\{X(s)\} = \mathcal{L}^{-1}\{2 + \frac{3}{s+1} - \frac{5}{s+2}\}$

$2\delta(t) + 3\cdot e^{-t}\cdot u(t) - 5\cdot e^{-2t}\cdot u(t)$

sağ taraflı, sağ taraflı

$2s^2+4s+5 \qquad s^2+3s+2 \qquad \frac{2s^2}{s^2+3s+2}$

Örnek: $H(s) = \frac{1}{s^2 - 1}$, $-1 < Re(s) < 1$

nedensel değildir

Örnek: $H(s) = \frac{e^s}{s+1}$, $Re(s) > -1$

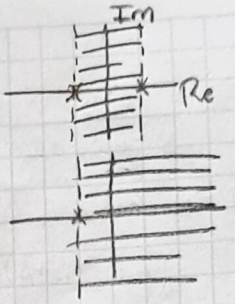
1. şart

$e^s \cdot \frac{1}{(s+1)}$

$\mathcal{X}\, h(t) = e^{-t} \cdot u(t)$

$\Rightarrow h(t) = e^{-(t+1)} \cdot u(t+1)$

Nedensel değil



2. şart rasyonellik

rasyonel ise nedensel olabilir

değilse nedensel değildir.

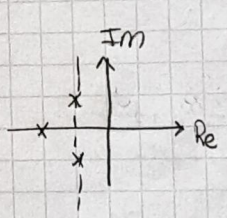
BIBO Kararlılığı

$H(s)$'in bütün kutupları sol düzlemde olmalı

Örnek: $H(s) = \frac{1}{(s+2)(s^2+2s+2)}$

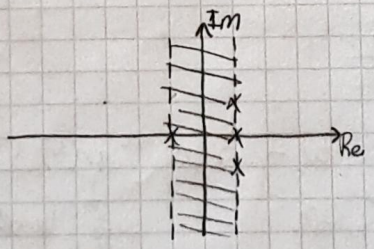
$S_1 = -2 \quad S_3 = -1 - j$

$S_2 = -1 + j$



Örnek: $\frac{s}{(s+1)(s-1)(s-1+j)(s-1-j)}$ $Re(s)$

?



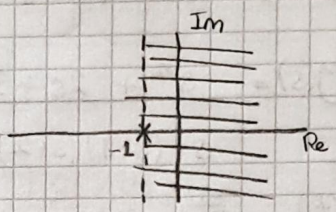
! En genel ifade olarak; ROC_H sanal ekseni kapsıyorsa H sistemi BIBO kararlıdır.

0'da bir kutup olsaydı kararlı olmazdı çünkü sanal ekseni kesmezdi

Tersine Çevrilebilirlik

$$H_{inv}(s) = \frac{1}{H(s)}$$

Örnek: $H(s) = \frac{s+1}{s+2}$, $Roc(s) > -2$



$H_{inv}(s) = \frac{s+2}{s+1}$

1) $Re(s) > -1$

2) $Re(s) < -1$

Örnek: $y'(t) + 3y(t) = x(t)$, $x(t) = e^{-t}u(t)$, $y(0) = 1$

$y(t) = ?$

çift taraflı , $\frac{d\,y(t)}{dt} \overset{LT}{\longleftrightarrow} s \cdot y(s)$

tek taraflı , $\frac{d\,y(t)}{dt} \overset{LT}{\longleftrightarrow} sy(s) - y(0)$

$$s \cdot y(s) - \underbrace{y(0)}_{1} + 3 \cdot y(s) = X(s)$$

$$y(s)[s+3] = \frac{1}{s+1} + 1 = \frac{s+2}{s+1} \Rightarrow y(s) = \frac{s+2}{(s+1)\cdot(s+3)}$$

$$= \frac{A_1}{s+1} + \frac{A_2}{s+3}$$

$$y(t) = \frac{1}{2}e^{-t} + \frac{1}{2}e^{-3t} \qquad t \geq 0$$

Örnek: $y'''(t) + 3y'(t) + 2y = x(t)$ $\quad x(t) = 5\,u(t)$

$y(0) = 1 \quad y'(0) = -1$

$s^2 y(s) - s\,y(0) - y'(0) + 3s\,y(s) - 3y(0) + 2y(s) = 5/s$

$y(s)[s^2 + 3s + 2] = \frac{5}{s} + s + 2$

$y(s) = \frac{s^2 + 2s + 5}{s \cdot (s+1)(s+2)} = \frac{A_1}{s} + \frac{A_2}{s+1} + \frac{A_3}{s+2} = \frac{5/2}{s} - \frac{4}{s+1} + \frac{3/2}{s+2}$

$y(t) = \frac{5}{2}u(t) - 4e^{-t}u(t) + \frac{5}{2}e^{-2t}u(t)$

Z Dönüşümü (Z Transform)

$z = e^s = \sigma + j\Omega = r \cdot e^{j\Omega}$ Fark özelliği

$X(s) = \int_{-\infty}^{\infty} x(t) \cdot e^{-st}\,dt$

$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$

Örnek: $x(n) = a^n \cdot u(n)$

$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n} \Rightarrow \sum_{n=0}^{\infty} a^n \cdot z^{-n} = \sum_{n=0}^{\infty} (a \cdot z^{-1})^n$

$\frac{1}{1-\alpha} = \frac{1}{1 - az^{-1}} = \frac{z}{z-a}, \; |z| > |a| \qquad a z^{-1} = \alpha$

Örnek: $x(n) = -a^n \cdot u(-n-1)$

$$X(z) = \sum_{n=-\infty}^{\infty} -a^n \cdot u(-n-1)\, z^{-n}$$

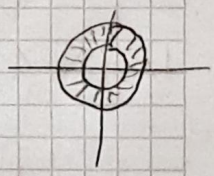
$$X(z) = -\sum_{-\infty}^{-1} a^n \cdot z^{-n} = -\sum_{n=1}^{\infty} z^n \cdot a^{-n}$$

$$= -\sum_{n=1}^{\infty} (a^{-1} \cdot z)^n = \frac{a^{-1} \cdot z}{1 - a^{-1} \cdot z} = \frac{z}{z-a} \qquad |z| < |a|$$

b) $x(n)$ sağ taraflı ise $X(z)_{ROC}$ bir çemberin dış bölgesi

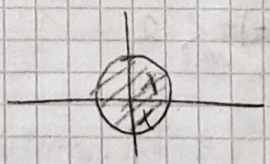
$x(n)$ sol taraflı ise $X(z)_{ROC}$ bir çemberin iç bölgesi

$x(n)$ çift taraflı ise $X(z)_{ROC}$ bir halkadır.

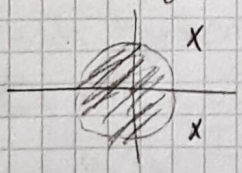


$x(n)$ sınırlı ve sonlu ise $X(z)_{ROC}$ tüm düzlem

Yakınsama bölgesi içinde kutup olamaz



Geçersiz



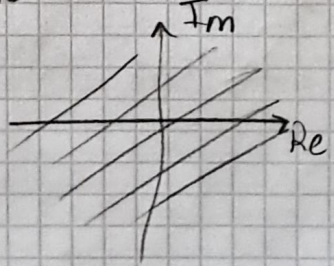
Geçerli

Örnek: $x(n) = \delta(n-n_0) \longleftrightarrow X(z)$, ROC_x

$$X(z) = \sum_{n=-\infty}^{\infty} \delta(n-n_0)\, z^{-n} = z^{-n}\Big|_{n=n_0}$$

$X(z) = z^{-n_0}$, $\forall z$,

ROC



Örnek: $x(n) = \frac{1}{n!} a^n \cdot u(n) \longleftrightarrow X(z)$, ROC

e^x Taylor açılımı

$$X(z) = \sum_{n=-\infty}^{\infty} \frac{1}{n!} \cdot a^n \cdot u(n) \cdot z^{-n} = \sum_{n=0}^{\infty} \frac{(a \cdot z^{-1})^n}{n!}$$

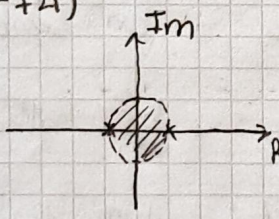
$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

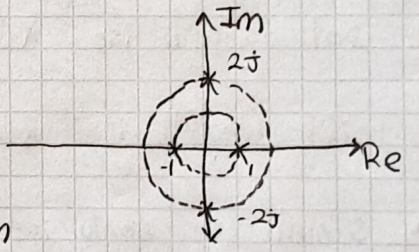
$$e^{a \cdot z^{-1}} \qquad \frac{a}{z} \rightarrow z = 0 \text{ tek kutup bölgesi}$$

$$|z| > 0$$

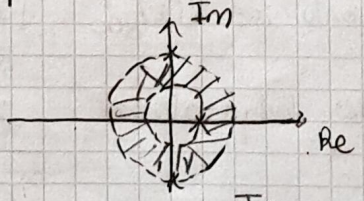
Örnek: $\frac{1}{(z^2-1)(z^2+4)}$ $z_1 = 1$ $z_2 = -1$ $z_3 = 2j$ $z_4 = -2j$

1) $|z| < 1$

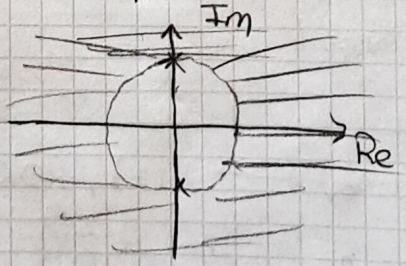




2) $1 < |z| < 2$



3) $|z| > 2$



Z Dönüşümünün Özellikleri

Doğrusallık Özelliği

Örnek: $x(n) = a^{|n|}$, $|a| < 1$

$$x(n) = x_1(n) + x_2(n)$$

$$x_1(n) = a^{-n}\, u(-n-1)$$

$$x_2(n) = a^n\, u(n)$$

$$X(z) = X_1(z) + X_2(z)$$

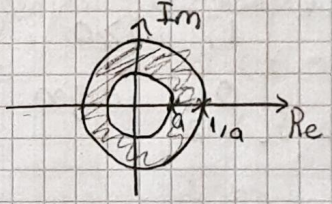
$$X(z) = \frac{z}{z - a^{-1}} + \frac{z}{z-a}$$

$$= \frac{(a - a^{-1})\cdot z}{(z-a)(z-a^{-1})}$$

$$\frac{z}{z - \frac{1}{a}}$$

$$X_1(z) = \frac{z}{z-a^{-1}}, \quad |z| < a^{-1}$$

$$X_2(z) = \frac{z}{z-a} \qquad |z| > a$$



Zamanda Kaydırma

Örnek: $x(n) = u(n-n_0)$

$$u(n) \overset{ZT}{\longleftrightarrow} \frac{z}{z-1} \qquad z > 1$$

$$u(n-n_0) \overset{ZT}{\longleftrightarrow} z^{-n_0}\cdot U(z)$$

$$= z^{-n_0}\cdot \frac{z}{z-1} \qquad |z| > 1$$

Z Domeninde Kaydırma

Örnek: $x(n) = a^n \cdot u(n)$

$$U(z) = \frac{z}{z-1} \overset{ZT}{\longleftarrow} u(n) \qquad ROC \quad z > 1$$

$$U(z/a) = \frac{z/a}{z/a - 1} \overset{ZT}{\longleftarrow} a^n \cdot u(n) \qquad ROC \quad z > a \cdot 1$$

Zamanda Ters Çevirme

Devamı 2 sayfa sonra

Örnek: $x(n) = u(-n)$

$$u(n) \xleftarrow{ZT} U(z) = \frac{z}{z-1} \qquad ROC\ |z| > 1$$

$$u(-n) \xleftarrow{ZT} U(1/z) = \frac{1/z}{1/z - 1} \qquad ROC\ |z| < 1$$

$$\frac{1}{1-z}$$

Upsampling

Örnek:

$$x(n) = \begin{cases} 1, & n \geq 0,\ n \text{ çift} \\ 0, & \text{diğer} \end{cases}$$

$$x(n) = (\uparrow 2)\, u(n) = u\left(\frac{1}{2}n\right)$$

$$u(z) = \frac{z}{z-1}, \quad M = 2, \quad |z| > 1$$

$$(\uparrow 2).\, u(n) \xleftrightarrow{ZT} u(z^2) = \frac{z^2}{z^2-1} = X(z)$$

Downsampling

Örnek: $y(n) = (\downarrow 2)\, x(n) \qquad M = 2$

$$X(z) = \frac{z^2-1}{z^2-\frac{1}{4}}, \quad |z| > 1/2, \quad y(z) = ?$$

$$y(z) = \frac{1}{2} \sum_{k=0}^{1} X\left(e^{-j2\pi k/2} \cdot z^{1/2}\right)$$

$$= \frac{1}{2} \sum_{k=0}^{1} X\left((-1)^k \cdot z^{1/2}\right)$$

$$= \frac{1}{2}\left[X(z^{1/2}) + X(-z^{1/2})\right]$$

$$y(z) = \frac{1}{2} \cdot \left[\frac{z-1}{z-1/4} + \frac{z-1}{z-1/4}\right] = \frac{z-1}{z-1/4} \qquad |z| > 1/4$$

Devamı 2 sayfa sonra

İşaretler ve Sistemler Dersi Ödevi B grubu

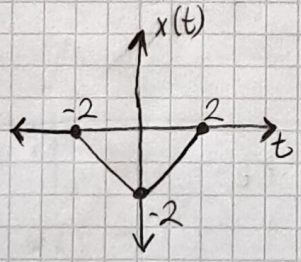
1. Soru

Bu sorunda verilen triangular fonksiyonudur ve tri(t) fonksiyonu şu şekilde ifade edilir.

$$tri(t) = \begin{cases} 1 - 2|t| & -0,5 < t < 0,5 \\ 0 & \text{diğer} \end{cases}$$

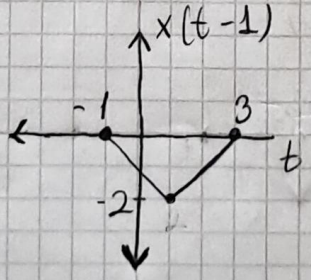
Burada verilen fonksiyonun genliği ve zamanı farklıdır.

a) $-2 \cdot tri(1/4\, t) = \begin{cases} -2 - t, & -2 < t < 0 \\ t - 2, & 0 < t < 2 \\ 0, & \text{diğer} \end{cases}$

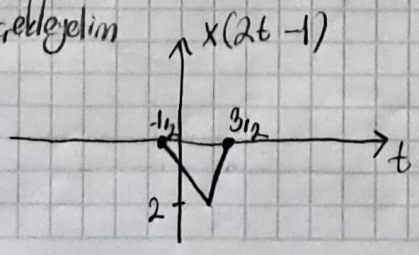


$x(2t-1)$ nedir? b)

Önce 1 birim sağa kaydıralım



Şimdi 2 ile ölçekleyelim

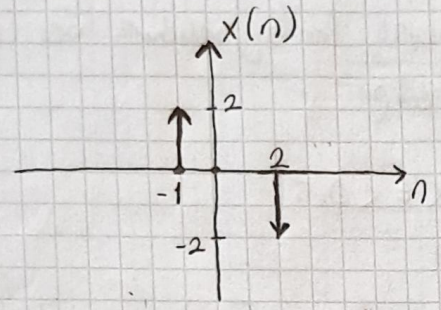


2. Soru

$x(n) = 2.[u(n+1) - u(n-2)]$

Yukarıda verilen sinyal ayrık bir sinyaldir ve birim dürtü tepkisidir.

a)



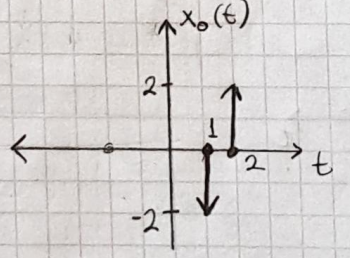
b)

$x_o(n) = -x(-n)$

$x_e(n) = x(-n)$

$x_o(n) = -2.[u(-n+1) - u(-n-2)]$

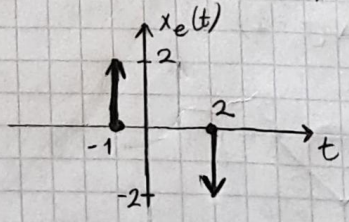
$= -2u[-n+1) + 2u(-n-2)$



$x_e(n) = 2[u(-n+1) - u(-n-2)]$

$= 2u(-n+1) - 2u(-n-2)$

$-(n-1)$ $n+2$



$\frac{2\pi}{\frac{\pi}{2}} = 4 \qquad \frac{2\pi}{\frac{3\pi}{2}} = \frac{4}{3}$

4. Soru

$$x(n) = 1 - 2\cos\left(\underbrace{\frac{\pi}{2}}_{T_1} n\right) + \cos\left(\underbrace{\frac{3\pi}{2}}_{T_2} n\right)$$

a) $\frac{T_1}{T_2}$ rasyonel ise periyodik

$T_1 = \frac{2\pi}{\omega_0} = \frac{2\pi}{\frac{\pi}{2}} = 4$

$T_2 = \frac{2\pi}{\frac{3\pi}{2}} = \frac{4}{3}$

$\frac{4}{\frac{4}{3}} =$ rasyonel

$\frac{T_1}{T_2} = 3 \qquad 3T_2 = T_1$

$\cancel{3} \cdot \frac{4}{\cancel{3}} = 4$

$T = 4$

3. Soru

$x(t) = e^{t+1} \cdot u(-t-1)$

$h(t) = u(t)$

$$x * h(t) = \int_{-\infty}^{\infty} x(\tau) \cdot h(t-\tau)\, d\tau$$

$x(\tau) = e^{\tau+1} \cdot u(-\tau-1)$

$h(t-\tau) = u(t-\tau)$

$$\Rightarrow \int_{-\infty}^{\infty} e^{\tau+1} \cdot u(-\tau-1) \cdot u(t-\tau)\, d\tau$$

Konvolüsyon

Örnek: $x(n) = u(n) * u(n)$

$$U(z) = \frac{z}{z-1}, \quad |z|>1$$

$$X(z) = U(z) \cdot U(z) = \frac{z^2}{(z-1)^2}, \quad |z|>1$$

z Domeninde Türev

Örnek $x(n) = \frac{1}{n!} \cdot n \cdot a^n \cdot u(n) \xleftrightarrow{ZT} X(z)$

$$x(n) = n \cdot V(n) \xleftrightarrow{ZT} -z \cdot \frac{d\,V(z)}{dz}$$

$$X(z) = -z \cdot \frac{d\, e^{a/z}}{dz} = -z \cdot \left(-a \cdot z^{-2}\right) e^{a/z}$$

$$X(z) = a \cdot z^{-1} \cdot e^{a/z} = \frac{a}{z} \cdot e^{a/z} \qquad |z|>0$$

Fark Alma

Örnek: $y(n) = x(n) - x(n-1)$ $\quad x(n) = a^n \cdot u(n)$, $y(z)$, ROC

$$X(z) = \frac{z}{z-a} \quad \xrightarrow{ROC} \quad |z|>a$$

$$y(z) = \frac{z-1}{z} \cdot \frac{z}{z-a} = \frac{z \cdot (z-1)}{z \cdot (z-a)} \quad \xrightarrow{ROC} \quad |z|>a$$

Biniştirme

$\delta(n) \xleftrightarrow{ZT} 1 = \delta(z)$

$x(n) = u(n)$ $\qquad u(n) = \sum_{k=-\infty}^{n} \delta(k)$

$X(z) = U(z) = \frac{z}{z-1}$ $\quad \delta(z) = \frac{z}{z-1}$ $\quad |z|>1$

Örnek: $X(z) = \frac{4z^2 - 3z}{2z^2 - 3z + 1}$ $\qquad x(0)$, $x(\infty) = ?$

$x(0) = \lim_{z\to\infty} X(z) = \lim_{z\to\infty} \frac{4z^2 - 3z}{2z^2 - 3z + 1} = 2$

$x(\infty) = \lim_{z\to 1} (z-1) \cdot X(z) = \lim_{z\to 1} \frac{(z-1)\cdot(4z^2 - 3z)}{2z^2 - 3z + 1} = 1$

Z Dönüşümü ödev

$x(n) = \sum_{k=-\infty}^{n} \left[u(k+5) - u(k) \right]$

$x(n) = n \cdot u(n-1)$

$x_2(n) = \left(\frac{1}{3}\right)^n u(n-1)$ $\qquad \sum_{n=-\infty}^{\infty} x(n) \cdot e^{-j\Omega n}$

$\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n \cdot e^{-j\Omega n} = \sum_{n=1}^{\infty} \left(\underbrace{\frac{1}{3} \cdot e^{-j\Omega}}_{\alpha}\right)^n$ $\qquad \frac{1}{1-\alpha}$

$\frac{1}{1 - \frac{1}{3} \cdot e^{-j\Omega}}$ $\qquad \frac{1}{3 - e^{-j\Omega}} = \frac{3}{3 - e^{-j\Omega}}$

$x_1(n) = (n+2) \cdot \delta(n+2)$ $\qquad \sum_{n=-\infty}^{\infty} x(n) \cdot e^{-j\Omega n} = \sum_{n=-2} (n+2) \cdot e^{-j\Omega n}$

$= 0$

Ters Z Dönüşümü

Örnek: $X(z) = \dfrac{1}{\left(z+\frac{1}{2}\right)\left(z-\frac{1}{2}\right)}$, $|z| > 1/2$

$$\frac{X(z)}{z} = \frac{A_1}{z+1/2} + \frac{A_2}{z-\frac{1}{2}} + \frac{A_3}{z}$$

$A_1 = 2$

$A_2 = 2$

$A_3 = -4$

$$\frac{X(z)}{z} = \frac{2\cdot 1}{z+1/2} + 2\cdot\frac{1}{z-1/2} - 4\,\frac{1}{z}$$

$$X(z) = \frac{2z}{z+1/2} + \frac{2\cdot z}{z-1/2} - 4\cdot 1$$

$$x(n) = 2\cdot\left(\frac{-1}{2}\right)^n\cdot u(n) + 2\cdot\left(\frac{1}{2}\right)^n\cdot u(n) - 4\cdot\delta(n)$$

Örnek: $X(z) = \dfrac{z\cdot(z+1)}{\left(z-\frac{1}{3}\right)\left(z-\frac{1}{5}\right)}$, $|z| > 1/3$ sağ

$$\frac{X(z)}{z} = \frac{A_1}{z-1/3} + \frac{A_2}{z-1/5} = \frac{-5}{z-1/3} + \frac{6}{z-1/5}$$

$$x(n) = -5\cdot\left(\frac{1}{3}\right)^n u(n) + 6\cdot\left(\frac{1}{5}\right)^n\cdot u(n)$$

Örnek: $X(z) = \dfrac{z^2\cdot(z-1)}{(z-1/4)(z-1/2)}$ a) $|z| < 1/4$ b) $\frac{1}{4} < |z| < \frac{1}{2}$ c) $|z| > 1/2$

$$\frac{X(z)}{z} = \frac{A_1}{z-1/4} + \frac{k_1}{z-1/2} + \frac{k_2}{(z-1/2)^2}$$

$$X(z) = \frac{-3z}{z-1/4} + \frac{4z}{z-1/2} - \frac{z}{\left(z-\frac{1}{2}\right)^2}$$

a) $|z| < 1/4$

$$x(n) = 3\cdot\left(\frac{1}{4}\right)^n\cdot u(-n-1) - 4\cdot\left(\frac{1}{2}\right)^n\cdot u(-n-1) - 2n\left(\frac{1}{2}\right)^n u(-n-1)$$

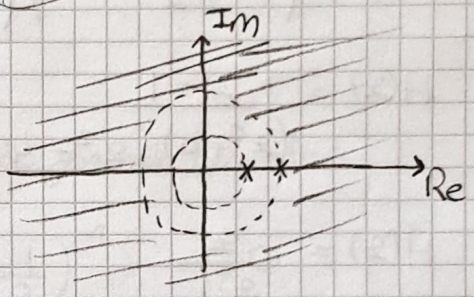
Örnek: $X(z) = \frac{z^3 + z^2 + 4z + 8}{z^2}$, $|z| > 0$

$$X(z) = z^1 + 1 + 4 \cdot z^{-1} + 8 \cdot z^{-2}$$

$$x(n) = \delta(n+1) + \delta(n) + 4 \cdot \delta(n-1) + 8 \cdot \delta(n-2)$$

Nedensellik

Örnek: $H(z) = \frac{2z^3}{\left(z - \frac{1}{4}\right)\left(z - \frac{1}{3}\right)}$, $|z| > 1/3$



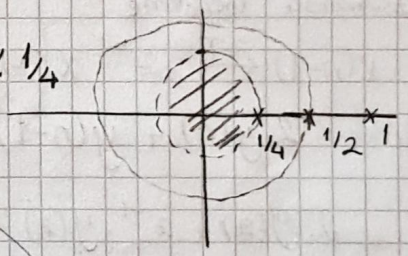
deg(N) > deg(D)

nedensel değil

BIBO Kararlılığı

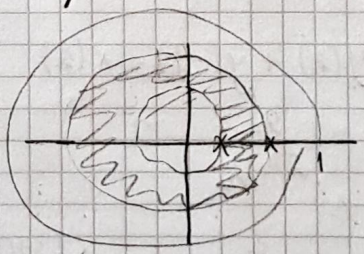
Örnek: $H(z) = \frac{z - 1/3}{(z - 1/4)(z - 1/2)}$

a) $|z| < 1/4$

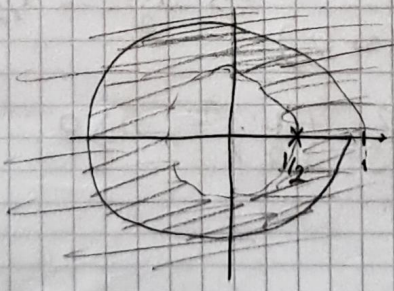


1'i kapsamıyorsa BIBO kararlı değildir

b) $\frac{1}{4} < |z| < \frac{1}{2}$



c) $|z| > 1/2$



Ödev

Örnek:

$$H(z) = \frac{z^2 - 1}{\left(z - \frac{1}{9}\right)\left(z^2 - \frac{1}{4}\right)}$$ nedensel, LTI

$$H(z) = \frac{z - 1}{z^2 - \frac{1}{4}}$$

Kararlılık için ROC = ?

$$H(z) = \frac{1}{\left(z - \frac{1}{2}e^{j\frac{\pi}{4}}\right)\left(z - \frac{1}{2}\cdot e^{-j\frac{\pi}{4}}\right)\left(z - \frac{3}{2}\cdot e^{j\frac{3\pi}{4}}\right)\left(z - \frac{3}{2}\cdot e^{-j\frac{3}{}}\right.}$$

$$H(z) = \frac{z - 1}{(z^2 + 4)(z^2 - 4)}$$ $e^{-i\pi} = -1$

$$H(z) = \frac{z}{z - 1}$$

Zamanda Gecikme

$x(n-1) \xleftrightarrow{ZT} z^{-1}X(z) + x(-1)$

Örnek: $2 \cdot y(n) + y(n-1) = x(n)$ $x(n) = \left(\frac{1}{4}\right)^n \cdot u(n)$, $y(-1) = 2$

$2 \cdot Y(z) + z^{-1} \cdot Y(z) + y(-1) = X(z)$ $y(n) = ?$

$$Y(z)\left[2 + \frac{1}{z}\right] = \frac{z}{z - 1/4} - 2$$ $X(z) = \frac{z}{z - 1/4}$

$$Y(z) = \frac{-z + 1/2}{z - 1/4} \cdot \frac{z}{2 \cdot (z + 1/2)} \Rightarrow \frac{Y(z)}{z} = 1/2\ \frac{-z + 1/2}{(z + 1/2)(z - 1/4)}$$

$$y(n) = \frac{1}{6}\left(\frac{1}{4}\right)^n - \frac{2}{3}\left(\frac{1}{2}\right)^n, \quad n \geq 0$$

Örnek $10y(n) + 3y(n-1) - y(n-2) = x(n)$

$x(n) = 6 \cdot u(n)$, $y(-1) = 3$, $y(-2) = 6$

$$10Y(z) + 3\cdot\left[z^{-1}\cdot Y(z) + y(-1)\right] - \left[z^{-1}\left(z^{-1}\cdot Y(z) + y(-1)\right) + y(-2)\right]$$
$$= X(z)$$

$$Y(z) = \frac{z(3z^2 + 6z - 3)}{(z-1)(10z^2 + 3z + 1)}$$
$$(5z-1)(2z+1)$$

$5\left(z + \frac{1}{5}\right)$ şeklinde yazmak iyi olur.

$$y(n) = \frac{1}{2} + \frac{3}{10}\left(\frac{1}{5}\right)^n - \frac{1}{2}\cdot\left(-\frac{1}{2}\right)^n, \quad n \geq 0$$

$$\left(\frac{3}{10}\cdot\frac{z}{z-1/5}\right) - \left(\frac{1}{2}\cdot\frac{z}{z+1/2}\right)$$